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FILAMENTOUS AND PLANTLIKE GREEN ALGAE

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I. INTRODUCTION

The green algae (Division Chlorophyta) possess chlorophyll *a* and *b* within a double membrane-bound chloroplast. True starch is stored in the chloroplast and forms the principal photosynthetic product. The cell wall is normally composed of cellulose, and flagellated stages, such as zoospores or gametes, are usually a part of the life history. Chlorophytes demonstrate a considerable amount of morphological variation, ranging from microscopic flagellated unicells to complex macroscopic thalli showing varying degrees of morphological differentiation. Traditionally the classification is based upon thallus organization with the green algae including a wide range of structural forms: single-celled and motile (flagellated), nonmotile (coccoid), colonial, filamentous, parenchymatous, and coenocytic (multinucleate). Considered in this chapter are those green algal orders in which the majority of the genera are filamentous (includes coenocytic forms), parenchymatous (tissue-like and often tubular or membrane-like in form) or possess a macroscopic plantlike thallus. Of the groups covered here, the Charales (charophytes) are the most plantlike and the order is sometimes considered in accounts of higher aquatic plants. The charophytes are commonly referred to as 'stoneworts' or 'brittleworts' since many are lime-encrusted ($CaCO_3$, especially *Chara*). Of the major orders of largely filamentous green algae the Zygnematales (reproduce by conjugation) is the only one excluded (see Chapter 9).

II. DIVERSITY AND MORPHOLOGY

The green algae contain over 500 genera and approximately 15,000 species (John, 1994), thus making it the largest algal division. The large majority (over 80%) live in freshwater with the remainder occupying habitats that include seawater/ brackish-water and soil or soil-free surfaces. Most are free living, but there are a few that are parasitic or symbiotic (Gartner, 1992; Chapman and Waters, 1992). In the orders Charales and Oedogoniales, almost all are confined to fresh water, unlike most other orders (e.g., orders Ulvales, Cladophorales) where the majority are marine.

A. MORPHOLOGY AND CELL STRUCTURE

Chloroplasts in the chlorophyte orders considered here vary greatly in size and shape. Commonly there is a single chloroplast lying against the inner cell wall and these vary from disk- or platelike, ringlike, or netlike (Hoek *et al*., 1995). There are many exceptions that include the Prasiolales, where the centrally positioned chloroplast is marginally lobed. Often the chlorophyll in terrestrial algae is masked by red carotenoid pigments (e.g., astaxanthin), which are sometimes dissolved in lipid droplets within the cell. Each chloroplast normally contains one to several pyrenoids (proteinaceous bodies) surrounded by a sheath of starch; pyrenoid number is a character of taxonomic importance. Its absence has to be viewed with caution, since they are known to arise *de novo* and are not always visible using the light microscope (LM). Taxonomically important characters are associated with ultrastructural features of the chloroplast and pyrenoid. Of the orders considered, only the Cladophorales, Siphonales, and Sphaeropleales are multinucleate. In the Siphonales, cross walls develop only during reproduction and thus the vegetative thallus is nonseptate (acellular) and multinucleate.

The cellulose cell wall of green algae has a structural fibrillar component and an amorphous matrix component; the latter is predominantly on the outer wall and often forms a slimy or mucilaginous outer layer. In a few orders the outermost layer is very hard and provides a suitable attachment surface for many algal epiphytes. Of the algae considered here, only the Trentepohliales and the Charales possess sporopollenin, a very hard and phenol-containing substance present in higher plant spore walls (Good and Chapman, 1978). Other green algae have decay-resistant wall compounds which differs chemically from sporopollenin in consisting of polymers of unbranched hydrocarbons and lacking phenolic groups (Gelin *et al*., 1997). Some green algae are characterized by having a diffluent or firm mucilaginous sheath or envelope.

B. REPRODUCTION

Vegetative reproduction normally takes place through fragmentation and simple cell division (Hoek *et al*., 1995). Asexual reproduction frequently involves biflagellate or quadriflagellate zoospores, except in the Oedogoniales where there is a ring of small cilia at the anterior pole. The cell functioning as a zoosporangium is similar in size to a vegetative cell or swollen and of a different form. Within a zoosporangium, one or more zoospores develop and are released through a pore(s) or by gelatinization of the wall. In some species, thick-walled, nonmotile spores, known as aplanospores, are produced. These have the ontogenetic possibility of developing into zoospores and thus are usually regarded as zoospores whose development has become arrested. Commonly formed are thick-walled resting vegetative cells known as akinetes; sometimes these are in series.

Sexual reproduction ranges from isogamy or anisogamy to oogamy (Hoek *et al*., 1995). In isogamy, the gametes are morphologically similar, and in anisogamy, the gametes differ in size; often gametes only fuse with those produced by a different individual. Oogamy is the most advanced form of reproduction, with the egg cell or oogonium retained and fertilized on the parent plant (e.g., Charales, Oedogoniales, Coleochaetales). Oogamy and other features (more than four meiospores per zygote) have led to the Coleochaetales being regarded as closely related to the ancestors of land plants (Graham and Wilcox, 2000). In some orders, the zygote resulting from fertilization develops a wall of sporopollenin whose ornamentation can provide taxonomically important characters (e.g., Charales, Leitch *et al*., 1990). For further information on reproduction in the chlorophytes, consult modern phycological textbooks such as Graham and Wilcox (2000) and Hoek *et al*. (1995).

Our knowledge of the life history of many green algae is still incomplete since ploidy levels of different stages are often unknown. In most of the green algal orders considered here, the only diploid stage is believed to be the resting zygote, a haplobiontic type of life history (John, 1994). Often these thick-walled spores require a period of obligate dormancy and this is considered an adaptation to freshwater life since many green algae have to survive seasonally adverse conditions. Only a few of the orders (e.g., Cladophorales, Trentepohliales) are known to have a regular alternation of diploid and haploid generations (diplohaplontic life cycles).

III. CLASSIFICATION OF GREEN ALGAE

The four genera of green algae (*Ulva*, *Conferva*, *Chara*, *Volvox*) mentioned in *Species Plantarum* (Linnaeus, 1753) were placed in the class Cryptogamia. It was not until Harvey (1836) created the Chlorospermeae that the green algae were recognized as a natural and separate assemblage. Pascher (1914, 1931) is credited with establishing thallus organization level as one of the most important characters for recognizing higher green algal groups. Since the 1950's three different approaches have contributed most to our understanding of green algal systematics and have led to a re-examination of phylogenetic relations. The most important resulted from ultrastructural research on cell division patterns (mitosis and cytokinesis) and flagellar organization. Cell division pattern and flagellar organization now provide the principal characters used for defining the Division (or Phylum) Chlorophyta, along with its classes and orders.

The first modern synthesis of new ultrastructural data and biochemical features led to the separation of two classes: Chlorophyceae *sensu* Stewart & Mattox and Charophyceae *sensu* Stewart and Mattox (1975; Pickett-Heaps, 1975). Later, two further classes were proposed, the Ulvophyceae (Stewart and Mattox, 1978) and Class Prasinophyceae (Mattox and Stewart, 1985). A fifth class has been recognized, the Pleurastrophyceae, but with the removal of the genus *Pleurastrum* it has been renamed the Trebouxiophyceae (Friedl, 1998). The five-class scheme has been tested several times and refined based primarily on further information on the ultrastructure of motile cells and analysis of molecular data (mostly small unit rDNA) (Friedl, 1998; McCourt, 1995; Booton *et al*., 1998). Two major lineages in the chlorophytes are accepted at present, one consisting mostly of microscopic freshwater forms and larger green marine algae (Chlorophyceae, Ulvophyceae, Trebouxiophyceae, Prasinophyceae), and the other lineage (class Charophyceae) of species more closely related to higher green plants than to other chlorophytes (Graham and Wilcox, 2000). The Charophyceae includes the Charales (charophytes) and the Coleochaetales; the latter order contains filamentous forms with an advanced form of oogamous reproduction and yet placed traditionally in the Chaetophorales.

The classification scheme adopted by Hoek *et al*. (1995) is based upon new molecular and ultrastructural data in addition to characters associated with reproduction, vegetative morphology, and biochemistry. Twelve new classes (including the Chlamydophyceae) were recognized of which the following contain filamentous (including siphonaceous), parenchymatous or "plantlike" green algae:

- Chlorophyceae (Oedogoniales, Chaetophorales)
- Ulvophyceae (Ulotrichales, as Codiolales; Ulvales)
- Cladophorophyceae (Cladophorales)
- Trentepohliophyceae (Trentepohliales)
- Zygnematophyceae (Zygnematales, Desmidiales)
- Klebsormidiophyceae (Klebsormidiales, Coleochaetales)
- Charophyceae (Charales)

The more soundly based green algal orders (e.g., Charales, Oedogoniales, Trentepohliales, Cladophorales, and Zygnematales) have been supported by molecular and ultrastructural data. Other orders defined largely on thallus organization have proved to be very heterogeneous so that very similar morphological forms may fall within different evolutionary lineages, for example, the Ulotrichales (unbranched filaments), Chaetophorales (branched, often heterotrichous), and Ulvales (parenchymatous), suggesting that similarity in thallus organization may be the result of convergent evolution. For example, genera traditionally assigned to the order Ulotrichales [e.g., *Ulothrix*, *Uronema*, *Klebsormidium* (= *Hormidium*)] belong to different orders and classes when ultrastructural characters are taken into account.

The various newly proposed schemes for the classification of the Division Chlorophyta (e.g., Hoek *et al*., 1995; Friedl, 1998; Graham and Wilcox, 2000) are not adopted here because the classification is still in a state of flux with no consensus likely to emerge until ultrastructural and molecular studies are undertaken on a considerably greater number and range of taxa. Many taxa are putatively placed and some will undoubtedly be transferred to other orders and classes when investigated using modern methods. Therefore, the position of genera for which there exists no information on cell division and/or flagellar organization must be considered uncertain (*incertae sedis*). For the purposes of this chapter, all genera are placed within the framework of a traditional classificatory scheme. It is tempting to adopt the solution suggested by Hoek *et al*. (1995), namely to assign a genus to the order or class to which they might appear to belong on the basis of light microscope observations. The problem of adopting such a solution has been addressed already, namely that organization level might give a misleading impression of relationships (see comments concerning genera traditionally placed within the Ulotrichales).

The traditional classification of the green algae adopted by Dillard (1989) and modified from that used by Prescott (1962), Bourrelly (1988), and others is followed. It has been decided not to recognize families since many of the same problems associated with orders apply equally to them. Genera are arranged alphabetically under each of the orders.

IV. ECOLOGY AND DISTRIBUTION

A. Aquatic Habitats

The majority of the filamentous and 'plantlike' green algae are aquatic and live attached to hard surfaces, such as rocks, man-made objects (plastic, glass, concrete channels, etc.), aquatic plants (flowering plants, mosses, macroalgae) and various animals (mollusks, turtles, fish). Frequently, larger filamentous forms are not only attached to submerged vegetation but entangled with it or else are free-floating on or just beneath the water surface. Water plants that frequently provide attachment surfaces for filamentous algae include

Najas, *Myriophyllum*, *Ceratophyllum*, rushes, and sedges. Many filamentous algae growing on water plants are also capable of growing on other surfaces, as has been shown to be the case when artificial surfaces have been placed in aquatic habitats and examined after being retrieved at different time intervals (John and Moore, 1985; Sabater *et al*., 1998). Associated with surfaces are diverse assemblages of attached or unattached microscopic forms that only become noticeable to the naked eye when present in vast numbers. Generally the assemblage is dominated by green algae or diatoms along with sessile and moving animals. The term "aufwuchs," or periphyton, is used for this assemblage of microscopic algae whether on submerged plants, rocks, or other surfaces (Chapter 2, Section II–F). The filamentous algae forming the periphyton include crustose forms (e.g., *Aphanochaete*, *Protoderma*) and others having erect and/or prostrate systems of branches (e.g., *Stigeoclonium*, *Chaetophora*, *Draparnaldia*). Some species perforate soft limestone and are partly (*Gomontia perforans*) or completely (*Gongrosira debaryana*) endolithic.

The majority of aquatic algae require hard attachment surfaces. Unconsolidated surfaces, such as mud, sand, or gravel, are normally too unstable for such forms. Some form mats overlying muddy bottoms and usually develop in deep or otherwise quiet water. For example, a band of *Dichotomosiphon tuberosa* is known from a depth of about 11–17 m in the Straits of Mackinac area of Lake Michigan and Lake Huron (Henson, 1984). The only green algal group well adapted to growing on beds of sand or silt are the charophytes or stoneworts. In marl-rich lakes or ponds recently cleared of higher plant vegetation, the charophytes may form extensive underwater "meadows". In such charophyte-dominated water bodies, plankton numbers tend to be low and the water very clear. If left unmanaged, small ponds containing charophytes become, in time, dominated by competitively superior aquatic macrophytes (Hutchinson, 1975).

One of the most common and important filamentous green alga in freshwater is *Cladophora*, commonly known as "cotton-mat" or "blanket weed" because dead and dying stranded masses turn gray or brown and resemble pieces of a woolen blanket (Canter-Lund and Lund, 1995; Chapter 24). These common names are usually taken as referring to mats of living *Cladophora*, but these are likely to contain and sometimes to be dominated by other filamentous green algae (particularly *Oedogonium* and *Spirogyra*; Chapter 9). Many secondarily detached green algae (including broken filaments) often become a nuisance when forming large free-floating mats in drainage ditches, ornamental ponds, and other small water bodies. *Cladophora* grows in greatest abundance along the shallow margins of nutrient-enriched lakes and on rocks in streams and faster-flowing rivers. *Cladophora glomerata* and the unbranched filaments of *Ulothrix zonata* are the dominant algae in the marginal shallows of nutrient-enriched lakes such as the Laurentian Great Lakes (Garwood, 1982) where *Cladophora* has become a serious problem alga. Dead and dying plants are known to accumulate in vast quantity along downwind shores and in sheltered bays thus reducing the amenity value of an area and profoundly affecting the local ecology. The blanket weed can also cause similar problems when it accumulates in the quiet backwater of lowland rivers. For a more comprehensive account of problems caused by filamentous and other algae, see Chapter 23.

Some filamentous algae, such as *Cladophora glomerata*, are reported to disappear from quiet backwaters during the warmer summer months and only continue to survive in fast-flowing reaches during this time. Its morphology depends on habitat conditions; plants growing in still water conditions are bushy since the branches are short and arise at obtuse angles, whereas in swiftly flowing water the filaments are long and branch only infrequently at acute angles. Filamentous algae, such as *Cladophora* and *Oedogonium*, often attain their greatest size in streams and fast-flowing rivers, possibly a reflection that in flowing water there is an accentuated gradient across the laminar boundary layer of water allowing for the rapid exchange of gases and nutrients (Whitford, 1960; Raven, 1992). An unusual growth form is reported from some lakes resulting from filaments growing out radially to produce a mossy ball (often 10 cm or more in diameter) commonly referred to as "*Cladophora*-balls" (Kindle, 1934; Daily, 1952).

Only a few filamentous green algae are endophytic and invade the tissues of water plants. One of the most common of these endophytes is *Chlorochytrium lemnae*, normally associated with the duckweed *Lemna*, but is also known in the tissues of *Ceratophyllum demersum*, *Elodea* spp., and certain liverworts (Hutchinson, 1975). Filamentous green algae occur on animals, notably mollusks, turtles, and fish. One of the most well known is *Basicladia* whose filaments form a feltlike covering on the backs of snapping turtles when present in abundance. Other green algae reported on turtles include *Rhizoclonium hieroglyphicum*, *Dermatophyton*, and *Cladophora* (Edgren *et al*., 1953); the latter genus and an unidentified chaetophoralean alga were discovered by Vinyard (1955) growing on salmon in Lake Texoma, Oklahoma. Some of those growing on turtles also occur on alligators which according to Prescott (1983) are often "…veritable algal gardens and offer a variety of species for the less timid collector".

Sometimes microscopic filamentous algae are swept into the plankton, otherwise only a few are to be regarded as truly planktonic. The only planktonic filamentous genus is *Helicodictyon*, which sometimes forms sulfur-yellow colored blooms during the autumn in the acid and slightly brown water of ponds and lakes on the coastal plain along the eastern seaboard of the United States (Whitford and Schumacher, 1969).

B. Soil and Subaerial Habitats

In terrestrial environments, algae grow on or within soil, rocks, stones, snow, animals, and plants. Filamentous algae sometimes develop on the surface of soil where they may form macroscopic growths when in quantity (e.g., *Fritschiella tuberosa*, *Klebsormidium flaccidum*, *Protosiphon botryoides*). Other soil-living algae have been discovered only when moist soil or soils mixed with nutrient media are cultured in the laboratory. Such algae are often defined upon characteristics observed in laboratory culture and frequently cannot be recognized by an examination of field samples [e.g., *Pseudoschizomeris*, *Filoprotococcus* (=*Trichosarcina*), *Hazenia*].

Damp stones, wooden fencing, greenhouse glass, roofing tiles, and tree bark are frequently discolored by growths of green algae forming the so-called subaerial or aero-terrestrial algal community. The almost ubiquitous powdery green layer, often referred to as the *Protococcus-Pleurococcus* community, is characterized by *Desmococcus* and *Apatococcus*. These two genera grow as solitary cells or small cell packets, although *Desmococcus* is capable of producing short filaments under very moist conditions; traditionally the genera are placed in the Order Chaetophorales. Another very characteristic genus on moist rock surfaces (often on steep cliffs), wood, tree trunks, leaves, and other subaerial surfaces is *Trentepohlia*. Its distinctive orange or reddish color and often feltlike growth readily distinguishes it from the *Protococcus–Pleurococcus* community. Subaerial algae sometimes grow in close association with fungi to form what is sometimes referred to as a "protolichen". *Trentepohlia* is a common algal component (phycobiont) of lichens, which consists of an intimate symbiotic association between a fungus and one or more algae or cyanobacteria. Lichen-algae are outside the scope of this chapter, although species belonging to some of the genera considered here (e.g., *Dilabifilum*, *Leptosira*, *Cephaleuros*) are known to be the algal partner of some lichens (Gartner, 1992). An unusual subaerial alga is *Trichophilus welcheri* whose filaments commonly grow amongst the hair scales of two- and three-toed sloths (*Bradypus*) in the rainforests of Central America (Thompson, 1972a). Often its hairs are pinkish in color due to the presence of another alga, *Cyanoderma bradypodis*.

In the humid tropics various filamentous algae are common on tree trunks and associated with leaves as epiphytes or endophytes. Some produce a red carotenoid pigment that masks the green color of the chlorophyll pigments when present in quantity. Most of these algae are epiphytic (e.g., *Trentepohlia*, *Phycopeltis*, *Stromatochroon*) or endolithic with a few presumed to be parasites or semi-parasites (Thompson and Wujek, 1997). One such parasitic species is C*ephaleuros virescens*, whose largely endophytic branches cause discoloration and death of leaf tissues in a wide range of tropical and subtropical plants (Chapman and Waters, 1992). It is a serious disease of certain commercial plants and causes the red rust of tea plants. *Phyllosiphon* is responsible for the yellow or red spots in the tissues of *Arisaema*, a vascular plant commonly known as Jack-in-the-Pulpit (Smith, 1950).

An extreme environment for algae is the surface of more or less permanent snowfields. The only filamentous green alga among other forms associated with this inhospitable environment is *Raphidonema* (Hoham, 1973). Often snow banks containing large quantities of algae are streaked green, yellow, or reddish depending on the dominant alga and the extent to which the green chlorophylls are masked by the red carotenoid pigments (see also Chapter 2; Section VI–B).

V. COLLECTION AND PREPARATION OF SAMPLES

Macroscopic growths of filamentous and parenchymatous algae, as well as charophytes, can be collected by hand from the littoral zone of lakes or from the surface if free-floating. In deeper water, snorkel or SCUBA-living is necessary unless a remote sampler is used, such as a grapnel or dredge. A weighted three- or four-pronged grapnel hook is ideal for sampling deeper-water charophytes. Small individual colonies or mats of softer filamentous algae need to be collected with a fine forceps and placed separately in vials or collecting jars. Often many interesting filamentous forms grow as a feltlike or fuzzy covering over stones, grasses, tree roots, and aquatic plants (e.g., *Myriophyllum*, *Najas*, *Potamogeton*, water lillies), including the submerged portions of emergent plants such as sedges or rushes. A small portion can be removed by means of a knife or scalpel and put in a suitable collecting vessel. Stones or other hard surfaces are best scraped to remove the surface film of algae. A simple and effective method of removing the algal film from rushes or sedges is to place the forefinger around the stem and to move it upward while at the same time pressing the thumbnail against the surface. Rotting pieces of wood or soft limestone rocks should be examined and sampled since they might contain algae that penetrate the surface.

Many methods that involve scraping a surface often fail to adequately sample those green algae forming minute crusts or having both prostrate and erect filaments. Often it is not easy to detect these algae while examining portions of water plants unless the underlying cells have died and become almost opaque. Endophytic algae, such as *Chlorochytrium*, are most readily detected in *Lemna* leaves that are dead or dying since its spots become more evident when the leaves are no longer green. *Cladophora* and *Oedogonium* have very hard walls and as a result are frequently heavily epiphytized by many minute filamentous forms. Glass bottles, pieces of glass, plastic bags, or similar transparent surfaces should be examined. On these artificial surfaces it is possible to detect very minute forms that are likely to be common on natural surfaces nearby. Of course, it is possible to provide substrata for colonization by these minute algae. Glass slides, plastic Petri dishes, or plastic bags are suitable surfaces and become colonized by such algae if left in the water for at least two weeks. It must be borne in mind that such artificial surfaces are selective and so not all small benthic forms might be sampled in this way.

Various algae growing on subaerial surfaces appear reddish or orange in color and are usually readily collected by scraping rock surfaces or by collecting discolored snow, especially from permanent snow banks. Subaerial algae are especially common in the humid tropics and grow as crusts, tufts, or mats on twigs, trunks, and leaves of trees and other plants. These algae should be collected along with the underlying surface to which they are attached. Various animals have filamentous or crustose algae growing on their shells and carapace (e.g., snapping turtles) and these usually have to be removed by a forceps or scraping. Sometimes soil algae form conspicuous surface mats if present in quantity. Otherwise, many soil algae can only detect and examined when soil is mixed with nutrients and cultured under laboratory conditions (“enrichment cultures”). Many of the filamentous soil algae mentioned are poorly defined and their taxonomy is based on features only observed in culture. Relatively few species of filamentous green algae are recorded from the plankton. Many of those recorded have been swept into the plankton by water turbulence. Such algae are collected along with truly planktonic forms in net tows.

Whenever possible, algae should be examined in the living state. Filamentous algae can often survive in reasonable condition for several days provided they are not kept in direct sunlight and in a reasonably large volume of water. Sterile specimens belonging to the Oedogoniales or other groups sometimes become fertile under such conditions. To investigate morphological variation and life history characters, it is desirable to grow them in laboratory culture. For long-term preservation, 4% formalin is ideal, although FAA (formalin: acetic acid: ethyl alcohol) is considered more suitable for flagellated stages since flagella are less likely to be lost than when formalin is used alone. If freshly collected or preserved macroscopic forms are to be sent through the mail, all surplus liquid should be drained and the specimen placed in a securely sealed polyethylene bag. Such forms can be preserved by drying on herbarium sheets, with muslin or cloth used to prevent them adhering to the drying papers. Charophytes are relatively coarse and do not generally adhere to drying papers and thus have to be tied by thread or held by cloth tape to the herbarium sheets. Often care is needed when drying since heavily $CaCO_3$-incrusted charophytes readily disintegrated if too much pressure is applied to a herbarium press.

VI. KEY AND DESCRIPTIONS OF GENERA

These dichotomous generic-level identification keys to filamentous and “plantlike” green algae are artificial and based upon morphological characters observed with the light microscope; only when absolutely necessary are reproductive, ecological, or more obscure characters used. Some characters are not absolute (e.g., free-living or attached, solitary, or colonial), making it necessary on occasion to “back track” and follow an alternative proposition. There are genera for which a single character applies only to some of its species and for this reason a genus might appear two or three times in the key (indicated as “in part”). The use of special descriptive terms is kept to a minimum. All generic determinations arrived at using this key need to be confirmed by reference to detailed descriptions and illustrations (see “Guide to the Literature for Species Identification”).

A. Key to Groups*

1a. Macroscopic and consisting of whorls of branches bearing divided or undivided whorls of branchlets................................GROUP I

1b. Microscopic, or if macroscopic then of another form...2

2a. Parenchymatous, forming tubular branched thalli or flat plates of cells, often membrane- or bladelike.............................GROUP II

2b. Unicellular or filamentous..3

3a. Unicellular (sometimes form 2- or 4-celled packets), associated with surfaces or within tissues of plants or animals, sometimes growing together but each cell with its own attachment...GROUP V**

3b. Filaments simple or branched 4

4a. Filaments unbranched or possess rhizoidal lateral branches, uniseriate or rarely multiseriate above............................GROUP III***

4b. Filaments branched, often a prostrate and an erect system present, always uniseriate. ....GROUP IV

* *Elakatothrix* does not fall into any of the broad groups; rows of separate or loosely contiguous cells form "pseudofilaments" lying within a broad mucilaginous envelope (often planktonic).

** Unicellular algae included because traditionally placed in orders where most other genera are filamentous.

*** Some genera of zygnematalean algae are unbranched and filamentous although with very distinctive chloroplasts and reproduce by conjugation (see Chapter 9).

B. Key to Genera by Group

1. *Group I* (Order Charales)

1a. Branches corticated, very rarely ecorticate throughout or partially corticate, with spine cells usually present; branchlets undivided and with bract cells at nodes (Fig. 9A)..*Chara*

1b. Branches usually ecorticate, without spines...2

2a. Branchlets undivided; single-celled outgrowths from outermost cells of the branch nodes (stipulodes), form a single ring, downward-pointing...*Lamprothamnium*

2b. Branchlet divided (sometimes only minutely at apex); outgrowths from outermost cells on the branch nodes absent.........................3

3a. Branchlets forked one or more times, each unicellular ray similar (Fig. 9B)..*Nitella*

3b. Branchlets divided into unequal, multicellular rays (Fig. 9C)...*Tolypella*

2. *Group II* (Orders Ulvales, Prasiolales, in part)

1a. Thallus tubular and hollow (Fig. 7A)...*Enteromorpha*

1b. Thallus flat and membrane-like...2

2a. Single-layered initially, multilayered at onset of reproduction, with cells in pairs or in larger groups and separated by mucilage; cells each with a stellate chloroplast; terrestrial (Fig. 7E)...*Prasiola*, in part

2b. Single-layered and parenchymatous; cells each with a cup-shaped chloroplast; aquatic (Fig. 7D)....................................*Monostroma*

3. *Group III* (Orders Oedogoniales, Sphaeropleales, Microsporales, Ulotrichales in part, Cylindrocapsales, Cladophorales in part).

1a. Filaments uniseriate initially and later becoming multiseriate..2

1b. Filaments uniseriate throughout..5

2a. Cells quadrate to shorter than broad where filaments uniseriate, each with a stellate chloroplast........*Prasiola* (*Rosenvingrella* stage)

2b. Cells cylindrical where filaments uniseriate, each with a broad parietal and band-shaped chloroplast which sometimes almost encircles cell...3

3a. Cells quadrate and bricklike where filament multiseriate, thick-walled, each containing several pyrenoids; aquatic (Fig. 7C) ....*Schizomeris*

3b. Cells spherical or in sarcinoid groups where multiseriate, thin-walled, each with a single pyrenoid; soil environments......................4

4a. Filaments mostly uniseriate, only final few cells in more than one series and spherical in shape (Fig. 4P)................*Pseudoschizomeris*

4b. Filaments more extensively multiseriate, often cells angular and grouped in packets (sarcinoid) (Fig. 7B)..................*Filoprotococcus*

5a. Cells often having distally one or several ringlike growths or ridges (cap cells) (Fig. 8C)..*Oedogonium*

5b. Cells never possess ringlike growths or ridges..6

6a. Thallus consisting of very long tubular cells containing cytoplasmic units separated by vacuoles and bandlike chloroplasts (Fig. 6E)..*Sphaeroplea*

6b. Thallus distinctly filamentous..7

7a. Cell walls composed of two overlapping H-shaped sections, often apparent at end of a filament; pyrenoids absent (Fig. 4E) ....*Microspora**

7b. Cell walls never so constructed; pyrenoids present or absent...8

8a. Cells containing one or two protoplasts, with mucilage between protoplast and cross wall often with transverse lamellations (Fig. 4B)..*Binuclearia*

8b. Cells without separate protoplasts...9

9a. Chloroplast dense and netlike formed of interconnecting discs each with a pyrenoid...10

9b. Chloroplasts usually parietal and platelike, often with a single pyrenoid..12

10a. Filaments less than 40 µm in diameter; occasionally with 1- to 3-celled rhizoidal branches arising at right angles to main axis (Fig. 5A)...*Rhizoclonium*

10b. Cells usually more than 40 µm in diameter, without rhizoidal branches 11

11a. Cells less than twice as long as broad, cylindrical to barrel-shaped, slightly to markedly constricted at cross walls......*Chaetomorpha*

11b. Cells more than twice as long as broad, cylindrical, without constrictions at cross walls....................*Cladophora* (unbranched form)

12a. Filaments not enclosed within a mucilaginous sheath 13

12b. Filaments enclosed within a mucilaginous sheath 21

13a. Filaments readily fragmenting into short lengths (< 10 cells in number) or solitary cells...14

13b. Filaments not readily fragmenting..18

14a. Filaments forming zigzag growths on surfaces; cells spherical, oval or cylindrical with short, blunt apices (Fig. 4D) ....*Stichococcus*, in part

14b. Filaments not forming such a pattern; cells spindle-shaped, elongate cylindrical, or cylindrical with rounded apices.......................15

15a. Filaments readily dissociate into solitary, cylindrical cells with bluntly rounded apices (Fig. 4O)....................................*Gloeotilopsis*

15b. Filaments attached or dissociate into short lengths of filament with cells of a different shape...16

16a. Cells or short filaments attached by a small disc; on soils (Fig. 4N)..*Raphidonemopsis*

16b. Cells not attached, free-living; rarely associated with soil..17

17a. Cells spindle-shaped very elongated; aquatic and terrestrial (Fig. 4J)...*Koliella*

17b. Cells elongate-cylindrical and apices acuminate or gradually tapering to a point; terrestrial, only associated with snow (Fig. 4K) ....*Raphidonema*

18a. Chloroplast filling cells and structure indecipherable due to density of food storage (principally starch) material; cells ovoid and thick-walled; reproduction oogamous, with reproductive cells swollen and red or orange (Fig. 6I)...............................*Cylindrocapsa*

18b. Chloroplast and cells of another form; reproduction not oogamous and without swollen or colored reproductive organs..............19

19a. Filaments terminating in an asymmetrically pointed apical cell (Fig. 4A)...*Uronema*

19b. Filaments without a pointed apical cell 20

20a. Chloroplast parietal, platelike, usually lobed, partly (often > 80%) or fully encircling cell (Fig. 4F).......................................*Ulothrix*

20b. Chloroplast parietal, elliptical or platelike, not lobed marginally, normally encircling 80% or less of cell (Fig. 4C).....*Klebsormidium*

21a. Cells shorter than broad..22

21b. Cells longer than broad...23

22a. Cells ellipsoidal to subquadrate, always contiguous within an uninterrupted mucilaginous sheath; some species have two overlapping helmet-like pieces resulting in a rim around mid-region (Fig. 4G)..*Radiofilum*

22b. Cells oval or oblong, never with a rim, often an interrupted row of 2 or 4 cells within a series of loosely attached mucilaginous sheaths (Fig. 4M)........*Hormidiopsis*

23a. Cells in a contiguous row; pyrenoids absent (Fig. 4L)........*Gloeotila*

23b. Cells separated, often equidistant and in pairs; pyrenoids present (Fig. 4I)........*Geminella*

*Similar wall structure to the yellow-green genus *Tribonema* (see Chapter 11), but differs in several characters including having netlike rather than platelike chloroplasts

Group IV (Chaetophorales in part, Coleochaetales, Oedogoniales in part, Cladophorales in part, Siphonales in part)

1a. Filaments perforating limestone rock, wood, or mollusk shells (Fig. 1H)........*Gomontia*

1b. Filaments not perforating surfaces........2

2a. Filaments lime-encrusted 3

2b. Filaments not lime-encrusted 4

3a. Thalli of loose and often unilaterally branched filaments, with pairs of short, green cells alternating with longer almost colorless cells (Fig. 3F)........*Chlorotylium*

3b. Thalli of compact filaments and irregularly divided filaments, with all cells green and no such cellular differentiation (Fig. 3J) ....*Gongrosira*, in part

4a. Thalli tubular or threadlike and contain many nuclei (coenocytic), cross walls only associated with reproduction........5

4b. Thalli multicellular, with each cell uninucleate or multinucleate........6

5a. Terrestrial, parasitic in higher plants (especially members of family Araceae) (Fig. 6H)........*Phyllosiphon*

5b. Aquatic, free-living, repeatedly dichotomously divided with constrictions at base of each fork (Fig. 6F)........*Dichotomosiphon*

6a. Associated with soil, growing on or partly subterranean; terminal cells often conical........7

6b. Aquatic or subaerial; terminal cells rarely conical 8

7a. Filaments green above soil surface and colorless below; cells each having a single chloroplasts and terminal cells without a cap (Fig. 3H)........*Fritschiella*

7b. Filaments all usually on surface and only rhizoids sometimes colorless; cells having a netlike chloroplast and terminal cells sometimes having a distinctive cap (Fig. 8B)........*Oedocladium*

8a. Thalli as disklike expansions on or within the carapace of turtles........9

8b. Thalli disklike or otherwise, never associated with turtles........10

9a. Cells each possess a single nucleus and pyrenoid, rare on turtles (only *Protoderma involvens*) (Fig. 1G)........*Protoderma involvens*

9b. Cells each possess several nuclei and one to several pyrenoids, rare on surfaces other than turtles (Fig. 5D)........*Dermatophyton*

10a. Epizoic among the hair scales of the sloth (Fig. 1B)........*Trichophilus*

10b. Aquatic or terrestrial on vascular plants and soil-free surfaces........11

11a. Endophytic, parasitic, or semi-parasitic, within the intercellular spaces of vascular plants and giving rise to erect filaments bearing sporangia (Fig. 6A)........*Cephaleuros*

11b. Epiphytic, on other surfaces and of a different form than above........12

12a. Subaerial on trees and other terrestrial surfaces........13

12b. Aquatic........16

13a. Thalli grow within the substomatal cavity of vascular plants, consisting of a bulbose cell giving rise to a few-celled stalk (Fig. 6D)........*Stomatochroon*

13b. Thalli not within the substomatal cavity and of another form........14

14a. Closely adjoined filaments forming epiphytic disks on the leaves of higher plants (Fig. 6B)........*Phycopeltis*

14b. Felty mats or tufts of filaments with branches arising from middle of cell and at right angles to main axis, growing on vascular plants and other surfaces (Fig. 6C)........15

15a. Sporangia always solitary, globular to kidney-shaped, and papilla pore basal; gametangia sometimes lateral on the sporangium rather than apical; gametangia lateral or terminal........*Printzina*

15b. Sporangia often grouped, ovoid and papilla pore apical; gametangia always terminal (Fig. 6C)....................................*Trentepohlia*

16a. Thalli macroscopic, spherical, cushion-like, or of irregularly divided, nodulose, somewhat flattened branches formed of filaments embedded in a firm or soft mucilage (Fig. 3I)..*Chaetophora*

16b. Thalli not forming macroscopic growths with a definite outline..17

17a. Filaments closely aggregated and pseudoparenchymatous, at least in part..18

17b. Filaments free from one another throughout and openly branched..27

18a. Erect filaments absent and prostrate system forming a single- or multi-layered crust..19

18b. Erect and prostrate filaments, or only with erect filaments..23

19a. Cells each containing several chloroplasts; crusts multi-layered in center (Fig. 1A)...*Pseudulvella*

19b. Cells each containing a single parietal chloroplast; crusts single-layered throughout..20

20a. Bristles or fine hairs absent...21

20b. Bristles or hairs present...22

21a. Zoospores biflagellate (Fig. 1G)..*Protoderma*

21b. Zoospores quadriflagellate (Fig. 1E)...*Chamaetrichon*

22a. Bristles with a basal sheath, often sheath persists after loss of bristle (Figs. 2I, J)..*Coleochaete*, in part

22b. Bristles without a basal sheath (Fig. 2K)..*Aphanochaete*, in part

23a. Prostrate system concentric in appearance due to alignment of cross wall and chloroplasts of adjacent cells; erect system very reduced..*Stigeoclonium*, in part

23b. Prostrate system and erect system of another form...24

24a. Prostrate system multi-layered (cushion-like) and bears sparingly divided erect filaments; zoospores biflagellate............................25

24b. Prostrate system single-layered and bears a poorly to profusely developed erect system; zoospores quadriflagellate where known 26

25a. Pyrenoids evident; sometimes lime-encrusted (Fig. 3J)...*Gongrosira*, in part

25b. Pyrenoids absent (if present not visible with LM); never lime-encrusted (Fig. 1J)...*Leptosira*, in part

26a. Reproduction only by akinetes and aplanospores (Fig. 3K)..*Dilabifilum*

26b. Reproduction by zoospores...*Pseudendoclonium*

27a. Branches bearing bristles or hairs..28

27b. Branches without such projections..31

28a. Prostrate system only, consists of simple or irregularly divided filaments (Fig. 2K)...*Aphanochaete*, in part

28b. Erect system only...29

29a. Bristle formed as extension of lateral wall at distal end of cell, with no cross wall at base (Fig. 3E)...*Fridaea*

29b. Bristles not as above..30

30a. Less than six-celled and branches rudimentary or absent; minute epiphyte within mucilage of *Gloeotrichia* (cyanobacterium) (Fig. 2B)...*Thamniochaete*, in part

30b. More than six-celled and branching in one plane; never embedded in mucilage (Fig. 8A)..*Bulbochaete*

31a. Thalli consist of short filaments often bearing 1- or 2-celled short branchlets terminating in truncate cells; soil alga (Fig. 1I) ....*Hazenia*

31b. Thalli of another form; aquatic algae..32

32a. Filaments peculiarly twisted, indistinctly branched, enclosed within a tough oval or spherical mucilaginous envelope (Fig. 1M) ....*Helicodictyon*

32b. Filaments distinctly branched, without a mucilaginous envelope..33

33a. Freely branching erect system only and some genera attached by a disk-shaped holdfast or rhizoids...34

33b. Freely branching erect and prostrate system, or only a prostrate system..42

34a. Microscopic (<100 µm in height), often freely branched with cross wall distal to point of branching; pyrenoids absent (Fig. 3L) .... *Microthamnion*

34b. Macroscopic, coarse or delicate and gelatinous........35

35a. Thalli enclosed within a delicate mucilage envelope, with distinct main branches bearing oppositely, alternately, or whorls of narrower-celled branchlets........36

35b. Thalli without a mucilaginous envelope and branching of another form........37

36a. Cells of main branches similar throughout (Figs. 3C, D)........*Draparnaldia*

36b. Cells of main branches of two types: cylindrical or barrel-shaped and very short from which arise lateral branchlets (Fig. 3B) ....*Draparnaldiopsis*

37a. Thalli usually soft and main axes <50 µm in diameter; cells each having a parietal platelike chloroplast, pyrenoid and nucleus (Fig. 3A)........*Stigeoclonium*, in part

37b. Thalli usually coarse and main branches usually over 50 µm in diameter; cells each having a netlike chloroplast and many pyrenoids and nuclei........38

38a. Branching confined to base of main filaments; commonly on the backs of snapping turtles (Fig. 5F)........*Basicladia*

38b. Branching of another form; not growing on turtles........39

39a. Branches arising immediately below cross wall; akinetes rarely produced........40

39b. Branches sometimes arising a short distance below cross wall; akinetes common and large, thick-walled, intercalary, or terminal 41

40a. Cells of main axes considerable larger than those of short branchlets, walls not lamellate, each having a parietal band-shaped chloroplast and one or a few pyrenoids (Fig. 3G)........*Cloniophora*

40b. Cells of main axes and branchlets of similar size, walls thick and lamellated, each having a netlike chloroplast and large numbers of pyrenoids (Fig. 5C)........*Cladophora*

41a. Akinetes spherical and in series; inland saline environments (Fig. 5B)........*Ctenocladus*

41b. Akinetes oval or barrel-shaped, solitary; freshwater environments (Fig. 5E)........*Pithophora*

42a. Prostrate system only; minute endo- or epiphytes........43

42b. Prostrate and an erect system; never endophytic........44

43a. Filaments sometimes highly branched, with cell thin-walled and not bearing bristles; epiphytic on *Cladophora* and *Rhizoclonium* (Fig. 1K)........*Entocladia*

43b. Subfilamentous mass of 2–6 cells, with cells having thick, lamellated walls and each bearing 8–12 long, flexuous bristles; epiphytic on vascular aquatic plants (Fig. 2D)........*Polychaetophora*, in part

44a. Plants growing within mucilage of other algae; filaments bearing basally swollen hairs, or these arising laterally on short projections (Fig. 2F)........*Chaetonema*

44b. Plants not growing within mucilage of other alga; hairs developing terminally........45

45a. Prostrate system bearing 5- to 8-celled, simple erect branches tapering to fine points and slender, multicellular hairs or bristles (Fig. 2H)........*Pseudochaete*

45b. Prostrate and erect systems usually well-developed, sometimes branches giving rise terminally to long hairs (Fig. 3A) ....*Stigeoclonium*, in part

5. *Group V* (Chaetophorales, Siphonales in part)

1a. Endophytic, consisting of irregularly oval cells within the duckweed *Lemna* (Fig. 1N)........*Chlorochytrium*

1b. Free-living, aquatic or terrestrial........2

2a. Cells bearing simple or branched bristles or hairs; aquatic algae........3

2b. Cells without bristles or hairs; usually soil or subaerial algae........7

3a. Cells loaf-shaped, bearing 1–2 very fine, forked bristles (Fig. 2C)........*Dicranochaete*

3b. Cells of another shape, each bearing one or more unbranched bristles........4

4a. Cells bearing sheathed hairs........5

4b. Cells bearing unsheathed hairs..6

5a. Cells subglobose, depressed, embedded in mucilage and not connected by tubes, bearing a number of radiating bristles ....*Conochaete*

5b. Cells spherical, sometimes connected by mucilage tubes, each bearing a single, sheathed bristle (Fig. 2A).............*Chaetosphaeridium*

6a. Cell walls thick and lamellate, with 8–12 bristles arising from each cell (Fig. 2D)..*Polychaetophora*

6b. Cell walls thin, with 2–4 long, flexuose bristles arising from each cell (Fig. 2E)..*Oligochaetophora*

7a. Swollen, multinucleate vesicle growing on surface of moist soil, with downward-growing colorless rhizoids; soil habitats (Fig. 6G) ....*Protosiphon*

7b. Spherical and uninucleate cells, often as 2- or 4-celled cuboidal packets; subaerial or isolated from soil or stagnant water................8

8a. Isolated from surface of damp soil or stagnant water; cells with one or more pyrenoids (Fig. 1F)....................................*Pleurastrum*

8b. Growing as a green powdery layer on various terrestrial surfaces; cells with or without a pyrenoid..9

9a. Cells each with a small pyrenoid (Fig. 1C)..*Desmococcus*

9b. Cells without a pyrenoid (Fig. 1D)..*Apatococcus*

C. Descriptions of Genera (Figs. 1– 9[1])

Order Chaetophorales

Filaments are uniseriate, branched, consist of erect and prostrate systems of filaments, sometimes erect branches absent or very reduced and prostrate system pseudoparenchymatous and well developed, with or without hairs or bristles. Cells are uninucleate, each with a parietal chloroplast and one to several pyrenoids.

Apatococcus V. F. Brand (Fig. 1D)

Unicellular, cells often divided in two or three planes to form irregular, cuboidal packets, sometimes forming short uniseriate filaments. Cells each with a parietal and often lobed chloroplast, pyrenoids absent. Asexual reproduction by biflagellate and slightly flattened zoospores, 8–32(–64) produced per cell. Autospores spherical. Sexual reproduction is unknown.

Very common and widespread subaerial alga, usually on the bark of trees where it forms part of the ubiquitous *Protococcus-Pleurococcus* community and often confused with other members of it.

The current consensus is that there exists only a single very polymorphic species, whereas formerly several were recognized (Ettl and Gartner, 1995). Retained in the Chaetophorales where traditionally placed but on ultrastructural grounds it should be assigned to the Chlorococcales (Hoek *et al.*, 1995).

Aphanochaete A. Braun (Fig. 2K)

Filaments are creeping and uniseriate; unbranched or irregularly branched, with cells sometimes bearing on upper surface one to several, long, unicellular, basally-swollen bristles. Cells are cylindrical or inflated, each possessing a parietal, disc-shaped chloroplast, with one to several pyrenoids. Asexual reproduction is by quadriflagellate zoospores or aplanospores. Sexual reproduction is oogamous: oosphere is large with a stalk, male gametes are quadriflagellate. The zygote is thick-walled and contains reddish to yellowish oil droplets.

Aphanochaete is very widespread and a common epiphyte on submerged aquatic plants (e.g., *Myriophyllum*, *Sagittaria*, *Najas*, *Chara*) and filamentous algae (particularly *Oedogonium*, *Cladophora*, *Rhizoclonium*, *Vaucheria*), growing commonly in hard and often eutrophic water in a wide range of still and flowing-water habitats.

There are six species of which four are considered valid and the other two are doubtful (Tupa, 1974). The genus is in need of further revision.

Chaetonema Nowakowski (Fig. 2F)

Filaments are uniseriate and irregularly divided, with some short side branches perpendicular to main axis and arising from the middle of the cell, often terminating in a hair, sometimes basally swollen hairs borne on short branches. Cells are cylindrical, each possessing a parietal, platelike chloroplast encircling more than half of the cell, with one or two pyrenoids. Asexual reproduction is by two quadriflagellate zoospores produced in each cell, flagella of equal length. Sexual reproduction is oogamous: eight biflagellate male gametes are in each swollen gametangium and the female gamete is in an enlarged vegetative cell.

Chaetonema grows in association with the mucilaginous sheath of various freshwater algae, including *Draparnaldia*, *Chaetophora*, *Tetraspora*, and *Batrachospermum*. There are only two species, both of which occur in North America.

It has been suggested that *Chaetonema ornatum*

[1] All scale bars in the figures equal 10 μm unless otherwise indicated.

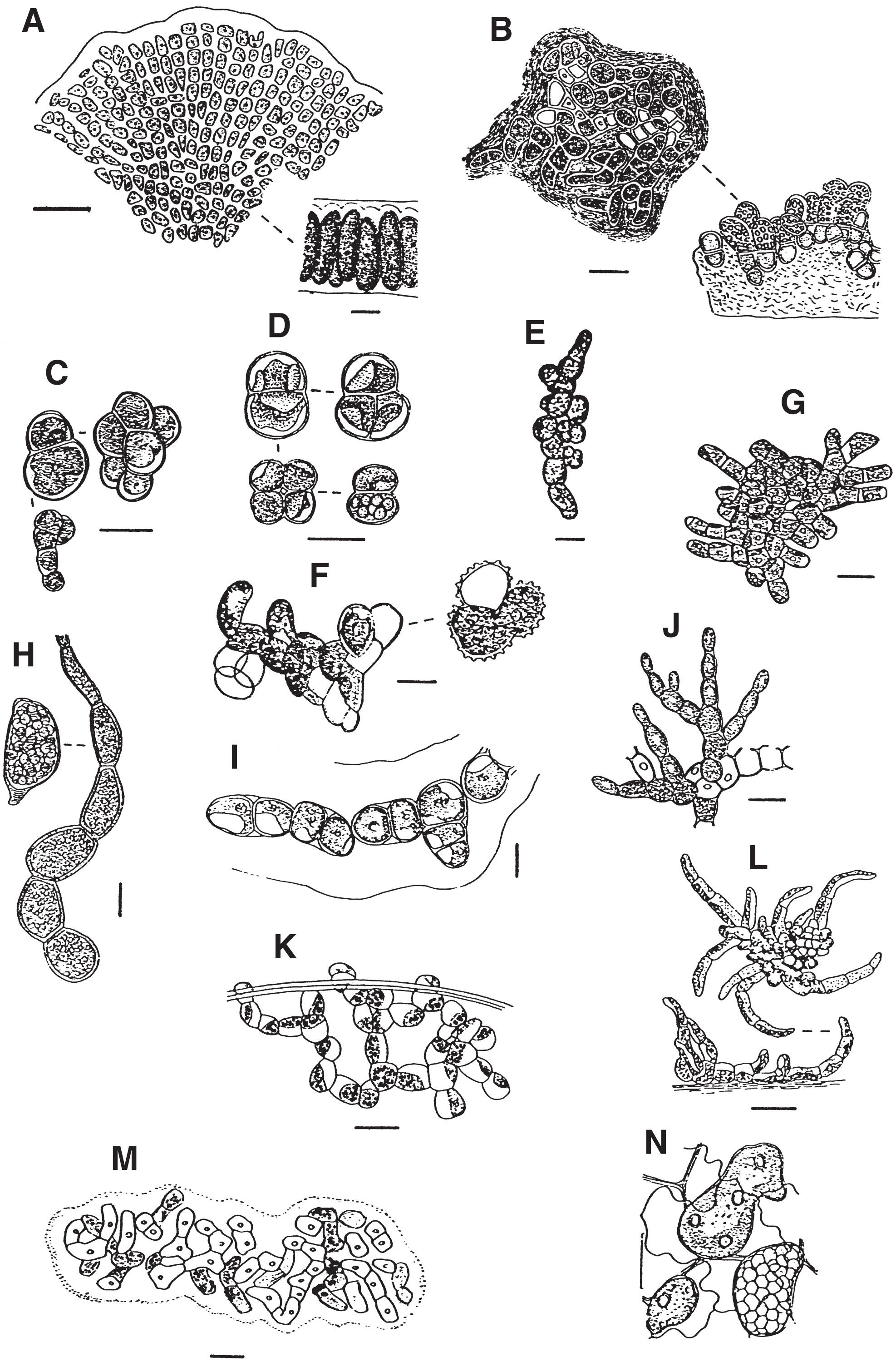

FIGURE 1 (A) *Pseudulvella americana*, surface view and section through thallus (after Snow, 1899). (B) *Trichophilus welcheri*, surface view and side view showing terminal sporangia (after Weber-van Bosse, 1887). (C) *Desmococcus olivaceus*, packets of cells and short filament (after Prescott, 1951). (D) *Apatococcus lobatus* (after Printz, 1964). (E) *Chamaetrichon capsulatum* (redrawn after Tupa, 1974). (F) *Pleurastrum insigne* (after Chodat, 1894). (G) *Protoderma viridae* (after Smith, 1933). (H) *Gomontia holdenii* (after Smith, 1933). (I) *Hazenia mirabilis* (after Bold, 1955). (J) *Leptosira mediciana* (after Borzi, 1883). (K) *Entocladia polymorpha* (after Prescott, 1951). (L) *Pseudendoclonium basiliense* (after Vischer, 1926). (M) *Helicodictyon planktonicum* (after Whitford, 1956). (N) *Chlorochytrium lemnae* (after Klebs, 1881).

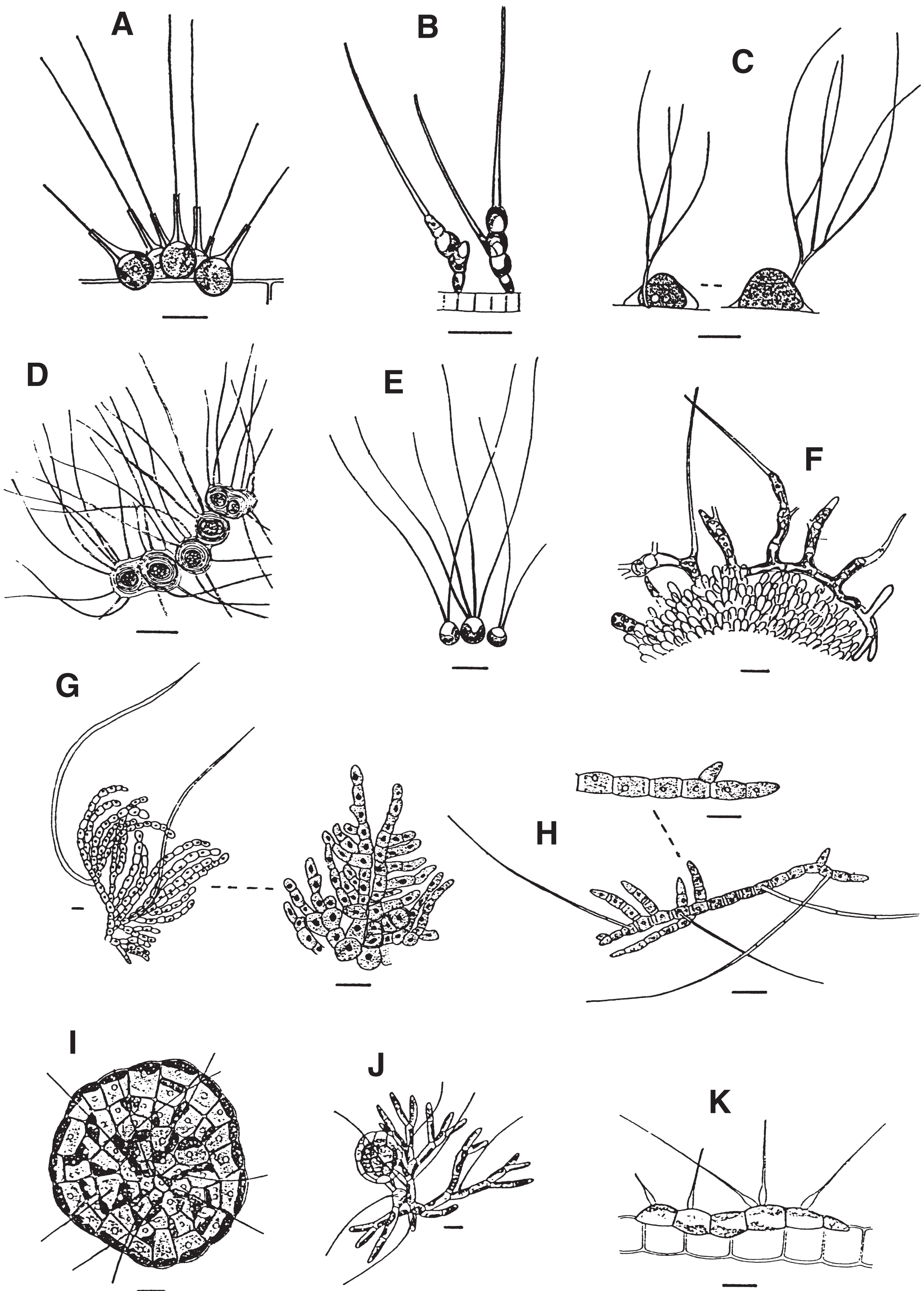

FIGURE 2 (A) *Chaetosphaeridium globosum* (after Smith, 1933). (B) *Thamniochaete huberi* (after Gay, 1891). (C) *Dicranochaete reniformis* (after Smith, 1933). (D) *Polychaetophora lamellosa* (after W. and G. S. West, 1903). (E) *Oligochaetophora simplex* (after West, 1911). (F) *Chaetonema irregulare* (after Huber, 1892). (G) *Trichodiscus elegans*, erect filaments with hairs and prostrate system without hairs (after Welsford, 1912). (H) *Pseudochaete gracilis* (after Smith, 1933). (I) *Coleochaete scutata* (after Smith, 1933). (J) *Coleochaete pulvinata* (after Smith, 1933). (K) *Aphanochaete repens* (after Smith, 1933).

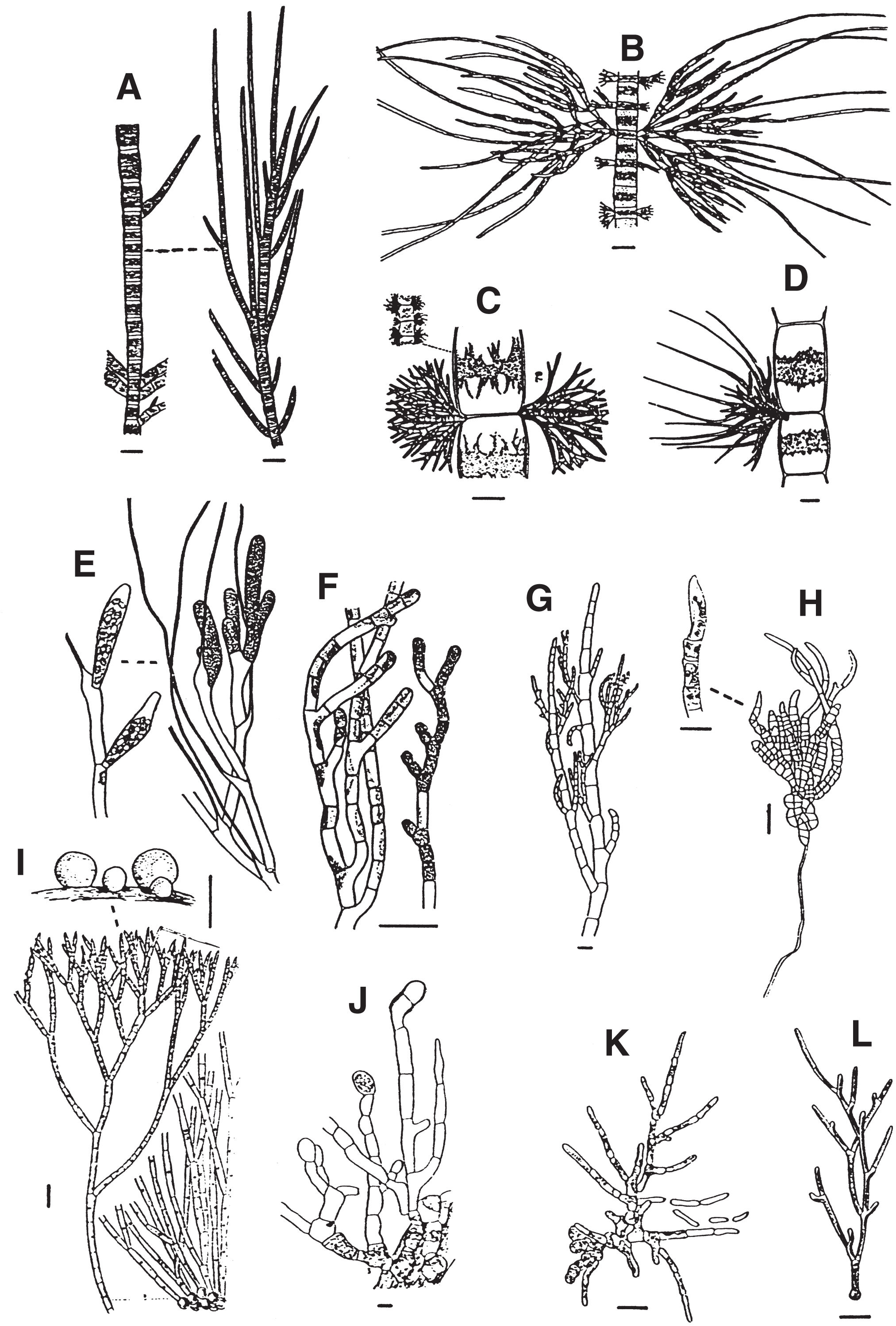

FIGURE 3 (A) *Stigeoclonium lubricum* (after Hazen, 1902). (B) *Draparnaldiopsis alpinis* (after Smith and Klyver, 1929). (C) *Draparnaldia ravenelii* (after Tiffany and Britton, 1952). (D) *Draparnaldia glomerata* (after Hazen, 1902). (E) *Fridaea torrenticola*, vegetative filament and one bearing bottle-shaped sporangia (modified after Smith, 1933). (F) *Chlorotylium cataractum*, vegetative filament and filament on right containing akinetes (after Smith, 1933). (G) *Cloniophora spicata* (after Prescott, 1983). (H) *Fritschiella tuberosa* (after Iyengar, 1932). (I) *Chaetophora elegans*, habit (after Prescott, 1951) and terminal and basal portions of colony (after Hazen, 1902). (J) *Gongrosira debaryana* (after Prescott, 1951). (K) *Dilabifilum printzi* (after Vischer, 1933). (L) *Microthamnion kuetzingianum* (after Tiffany, 1937).

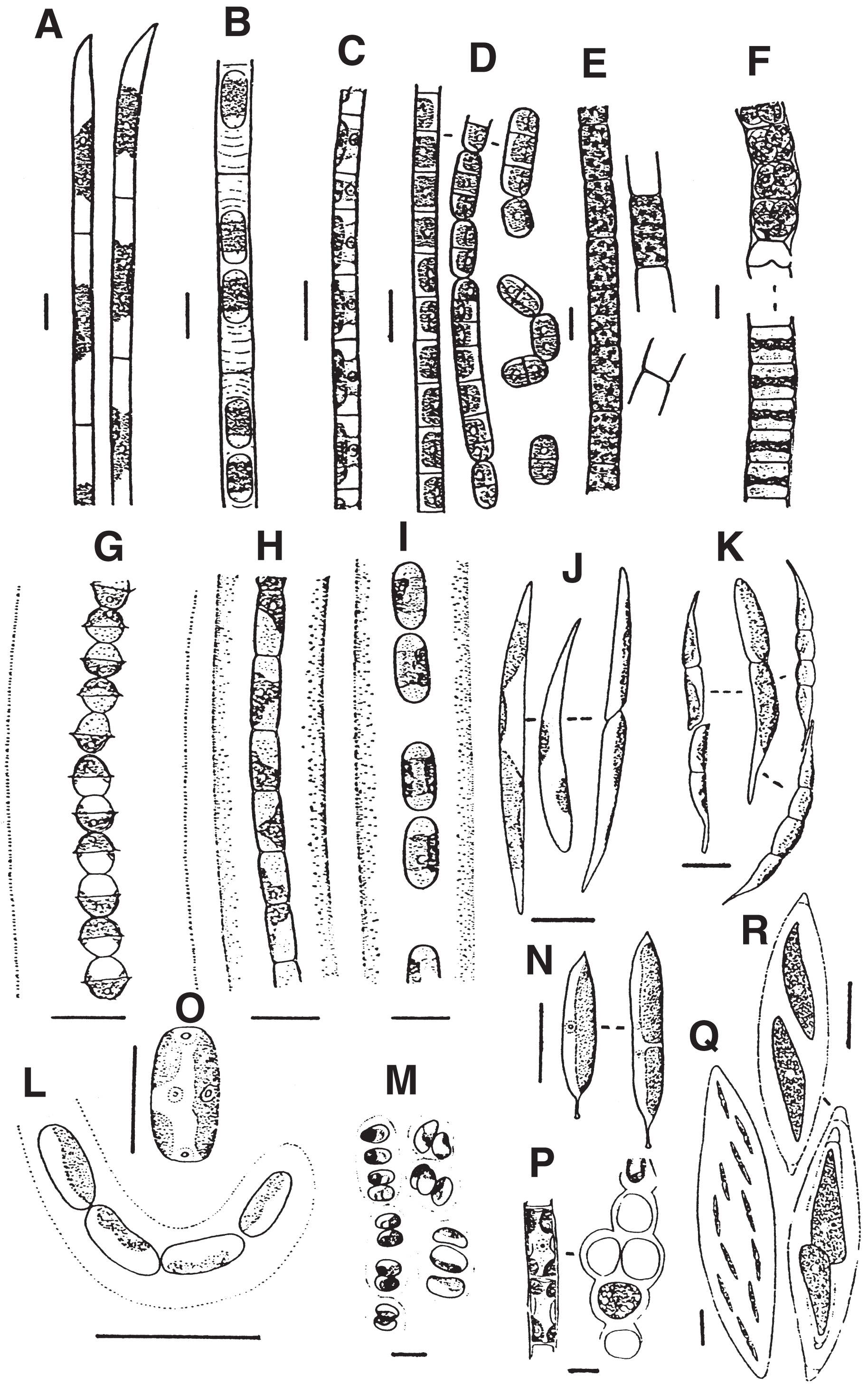

FIGURE 4 (A) *Uronema elongatum* (after Smith, 1933). (B) *Binuclearia tatrana* (after Smith, 1933). (C) *Klebsormidium klebsii* (after Smith, 1933). (D) *Stichococcus subtilis* (after Smith, 1933). (E) *Microspora willeana* (modified after Smith, 1933). (F) *Ulothrix zonata*, vegetative filament and one with the cells containing zoospores (after Smith, 1933). (G) *Radiofilum conjunctivum* (modified after Smith, 1933). (H) *Geminella minor*. (after Smith, 1933). (I) *Geminella interrupta* (modified after Smith, 1933). (J) *Koliella* sp. (after Prescott, 1983). (K) *Raphidonema nivale* var. *taylorii* (after Kol, 1938). (L) *Gloeotila contorta* (after Prescott, 1983). (M) *Hormidiopsis ellipsoideum* (after Prescott, 1983). (N) *Raphidonemopsis sessilis* (after Deason, 1969). (O) *Gloeotilopsis sterile* (after Deason, 1969). (P) *Pseudoschizomeris caudata*, uniseriate lower portion and multiseriate uppermost portion of filament (after Deason and Bold, 1960). (Q) *Elakatothrix gelatinosa* (after Smith, 1933). (R) *Elakatothrix viridis* Snow (after Smith, 1933).

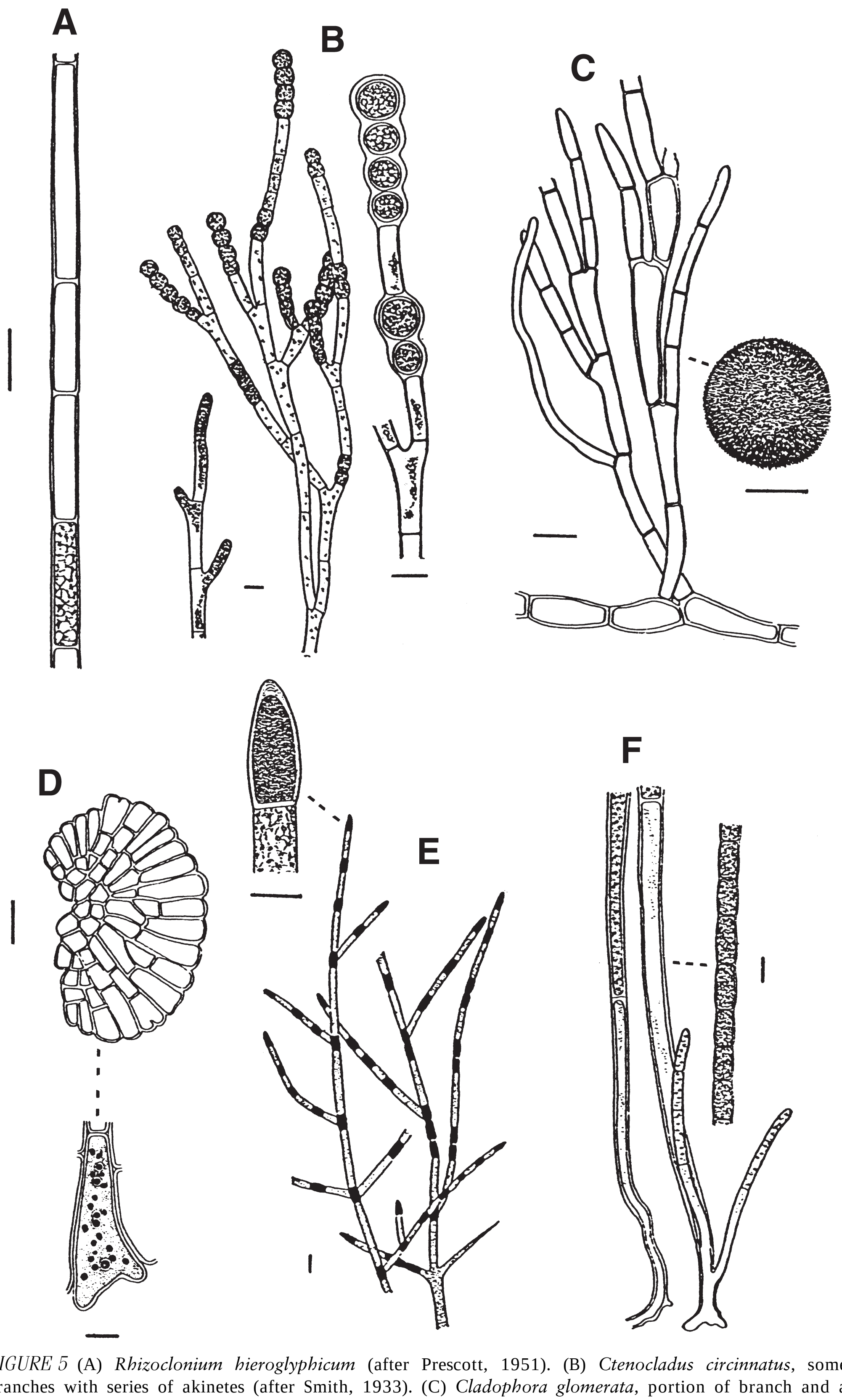

FIGURE 5 (A) *Rhizoclonium hieroglyphicum* (after Prescott, 1951). (B) *Ctenocladus circinnatus*, some branches with series of akinetes (after Smith, 1933). (C) *Cladophora glomerata*, portion of branch and a "*Cladophora*" ball (after Smith, 1933). (D) *Dermatophyton radians*, disklike thallus and close up of one of the multinucleate cells (after Feldmann, 1939). (E) *Pithophora oedogonia*, ortion of filament with akinete and close up of a terminal akinetes (after Smith, 1933). (F) *Basicladia chelonum* (after Smith, 1933).

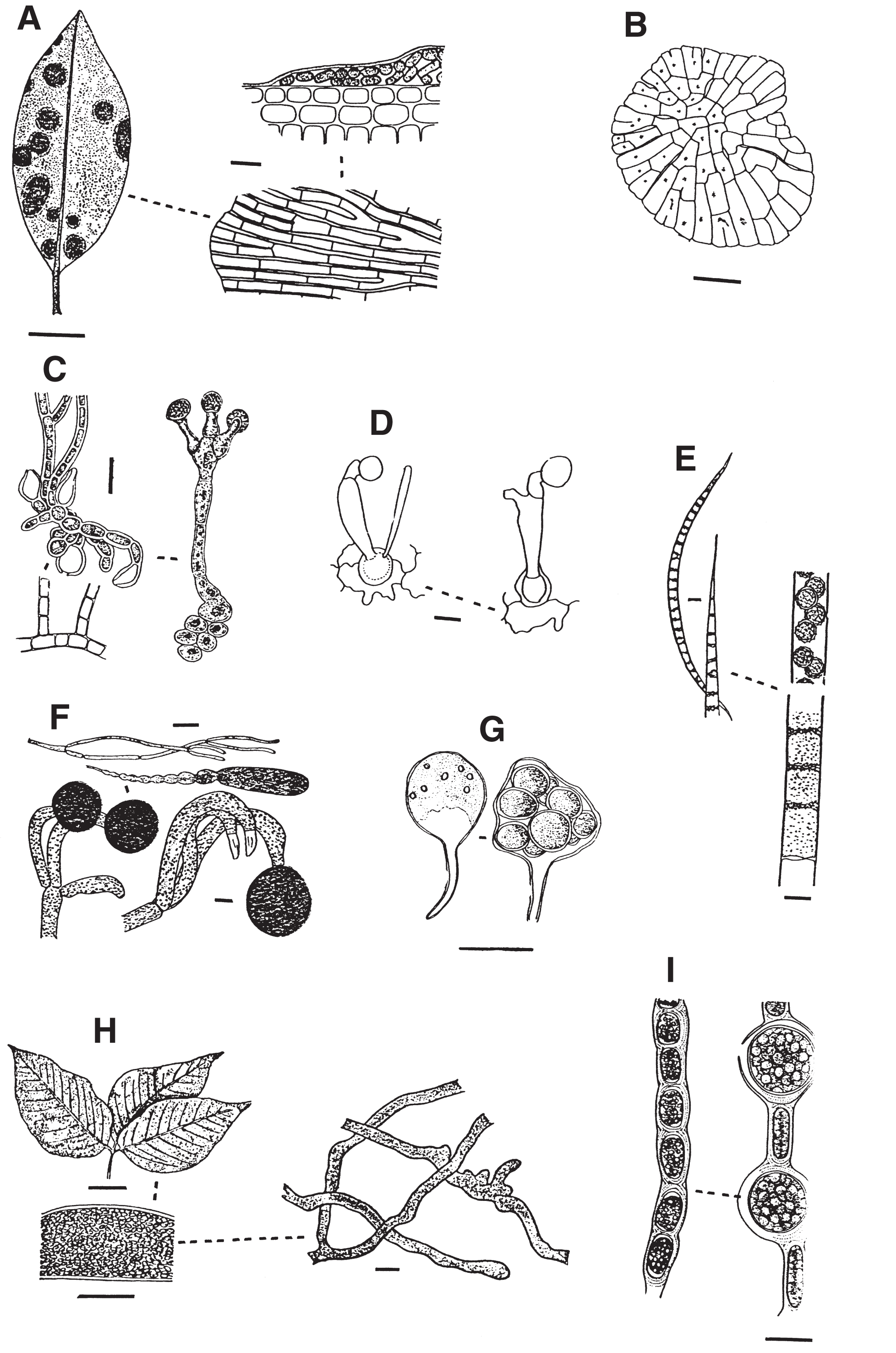

FIGURE 6 (A) *Cephaleuros virescens*, crusts of *Cephaleuros* on a *Magnolia* leaf and closer view of a crust and a section through a crust (after Smith, 1933). (B) *Phycopeltis arundinacea* (after Bourrelly, 1968). (C) *Trentepohlia aurea* (after Gobi, 1871). (D) *Stomatochroon lagerheimii* (after Palm, 1934). (E) *Sphaeroplea annulina*, portions of vegetative thallus and of one containing spores whose walls are thick and ornamented (after Prescott, 1951). (F) *Dichotomosiphon tubersosus*, tubular thallus and portions possessing sex organs (below left and right) as well an elongate akinetes (after Smith, 1933). (G) *Protosiphon botryoides*, young thallus and one containing thick-walled resting spores (after Bourrelly, 1967). (H) *Phyllosiphon arisari*, leaf of *Arisaema triphyllum* infected by *Phyllosiphon* and portions of thallus (after Smith, 1933). (I) *Cylindrocapsa geminella*, vegetative filaments and another containing oosporangia (modified after Prescott, 1951).

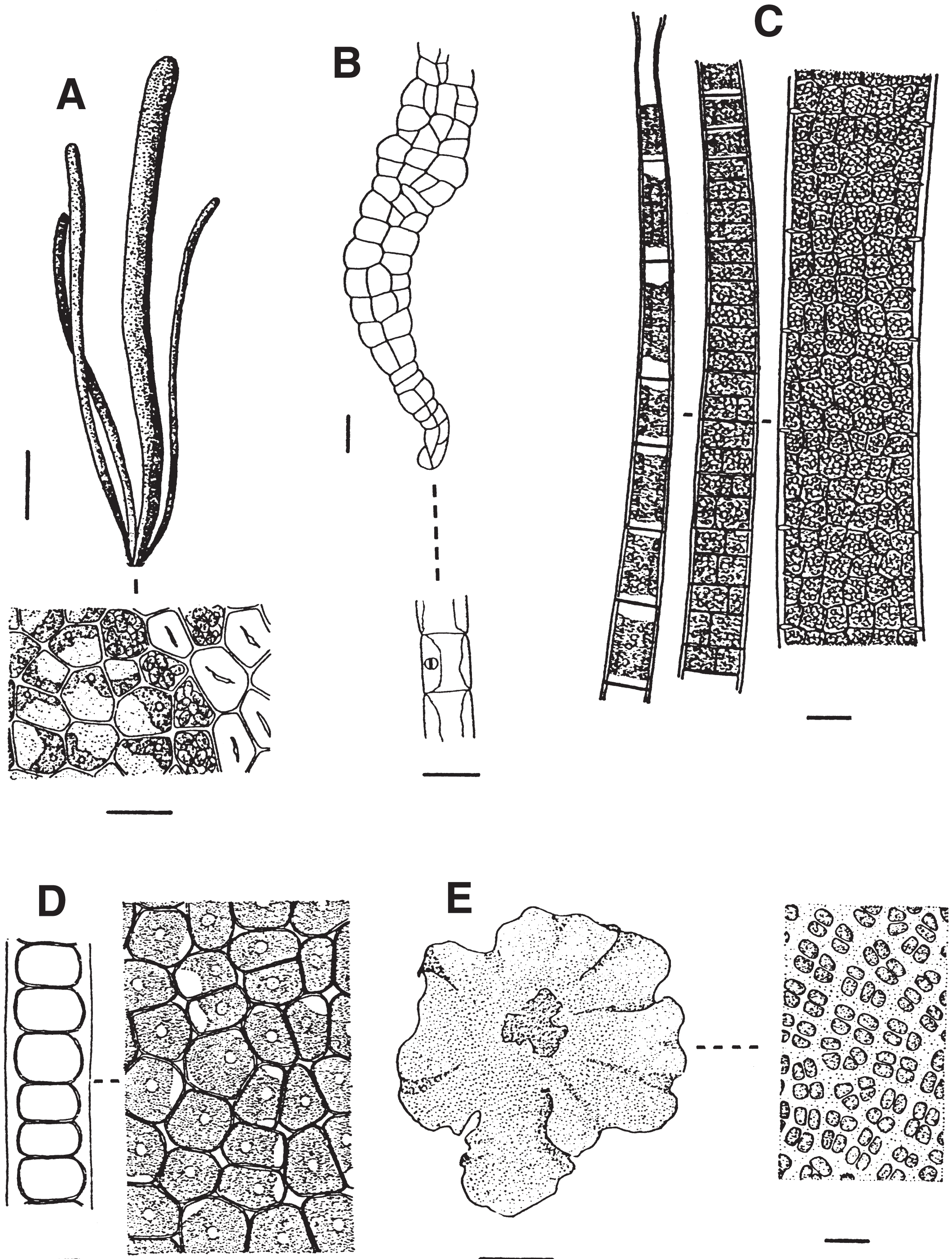

FIGURE 7 (A) *Enteromorpha intestinalis*, habit and section through tubular thallus (after Kützing, 1853–1856). (B) *Filoprotococcus polymorphum* (after Prescott, 1983). (C) *Schizomeris leibleinii* (after Smith, 1933). (D) *Monostroma latissimum* (after Smith, 1933). (E) *Prasiola mexicana*, habit and a portion of thallus (modified after Smith, 1933).

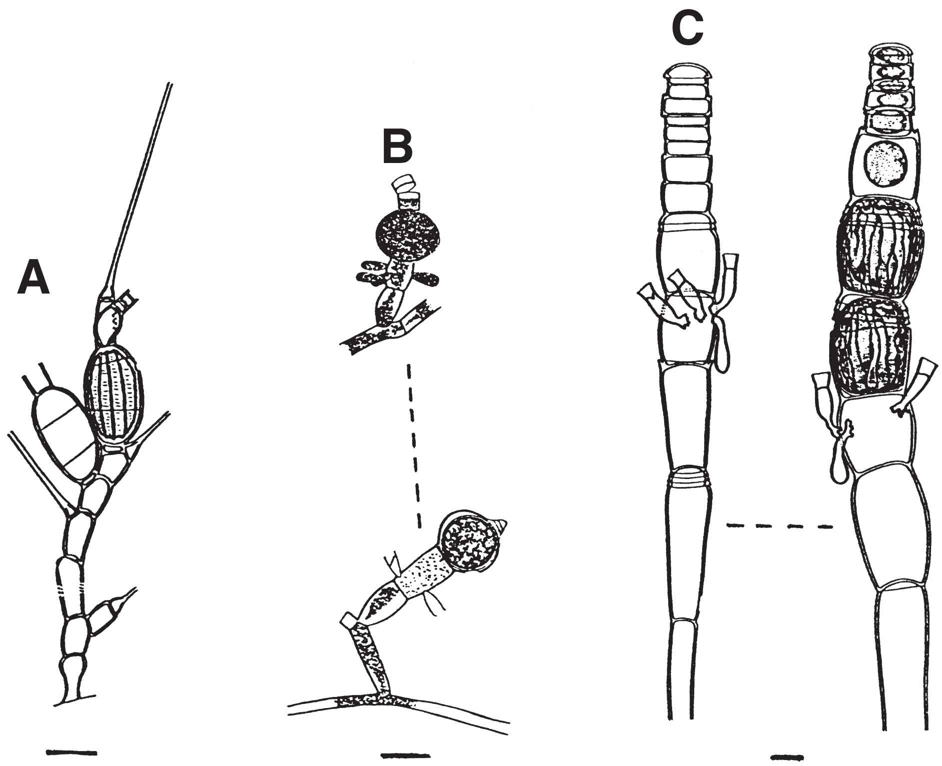

FIGURE 8 (A) *Bulbochaete minor*, portion of filament with swollen and ornamented zygospore (after Tiffany and Britton, 1952). (B) *Oedocladium hazenii*, portion of filaments with dwarf males and unfertilized oogonium (above) and of an oogonium containing a thick-walled zygospore (below) (after Smith, 1933). (C) *Oedogonium croasdaleae*, filament with dwarf males (left) and other filaments having dwarf males and two ornamented zygospores (right) (after Jao, 1934).

should be transferred to Thompson's doubtful genus *Chaetonemopsis* (Thompson, 1972b).

Chaetophora F. Schrank (Fig. 3I)

Chaetophora forms macroscopic growths, spherical, tubercular, arbuscular, or hemispherical enveloped in soft or firm mucilage, consisting of uniseriate filaments; a prostrate system of filaments is little-developed but erect system highly branched, intertwined filaments with each tapering to a blunt point or terminating in a long, multicellular hair. Cells contain a parietal chloroplast, sometimes bandlike, one to several pyrenoids. Zoospores are quadriflagellate; gametes are biflagellate and isogamous. Akinetes are generally produced in upper cells of branches and are brown in color.

Chaetophora grows on submerged aquatic plants (including tree roots) and a wide variety of other submerged surfaces in springs, pools, ponds, and lakes, as well as in streams and other relatively fast-flowing water courses. Many species are most abundant in winter and spring, evidently preferring cooler water conditions. Of the 12 species, at least four are known from North America.

Re-investigation is required as the species is distinguished principally on such unreliable criteria as the macroscopic form of the thallus and the nature of branching of the filaments forming it.

Chaetosphaeridium Klebahn (Fig. 2A)

The *Chaetosphaeridium* consists of spherical to flask-shaped cells, solitary or in dense groups, sometimes within a common mucilage envelope or connected in series by a tubular mucilaginous elongation, each cell bearing a long, simple, and prominently basally sheathed bristle. Cells contain one or two parietal, plate-like chloroplasts, and a single pyrenoid. Zoospores biflagellate, two or four per cell. Sexual reproduction is oogamous, oogonia are very swollen and male gametes biflagellate.

Chaetosphaeridium often grows on mosses and other submerged surfaces, rarely endophytic, frequently becomes dislodged and free-floating along with the plankton in ponds and creeks.

In most recent accounts, the genus is placed in the order Coleochaetales based on its ultrastructural features.

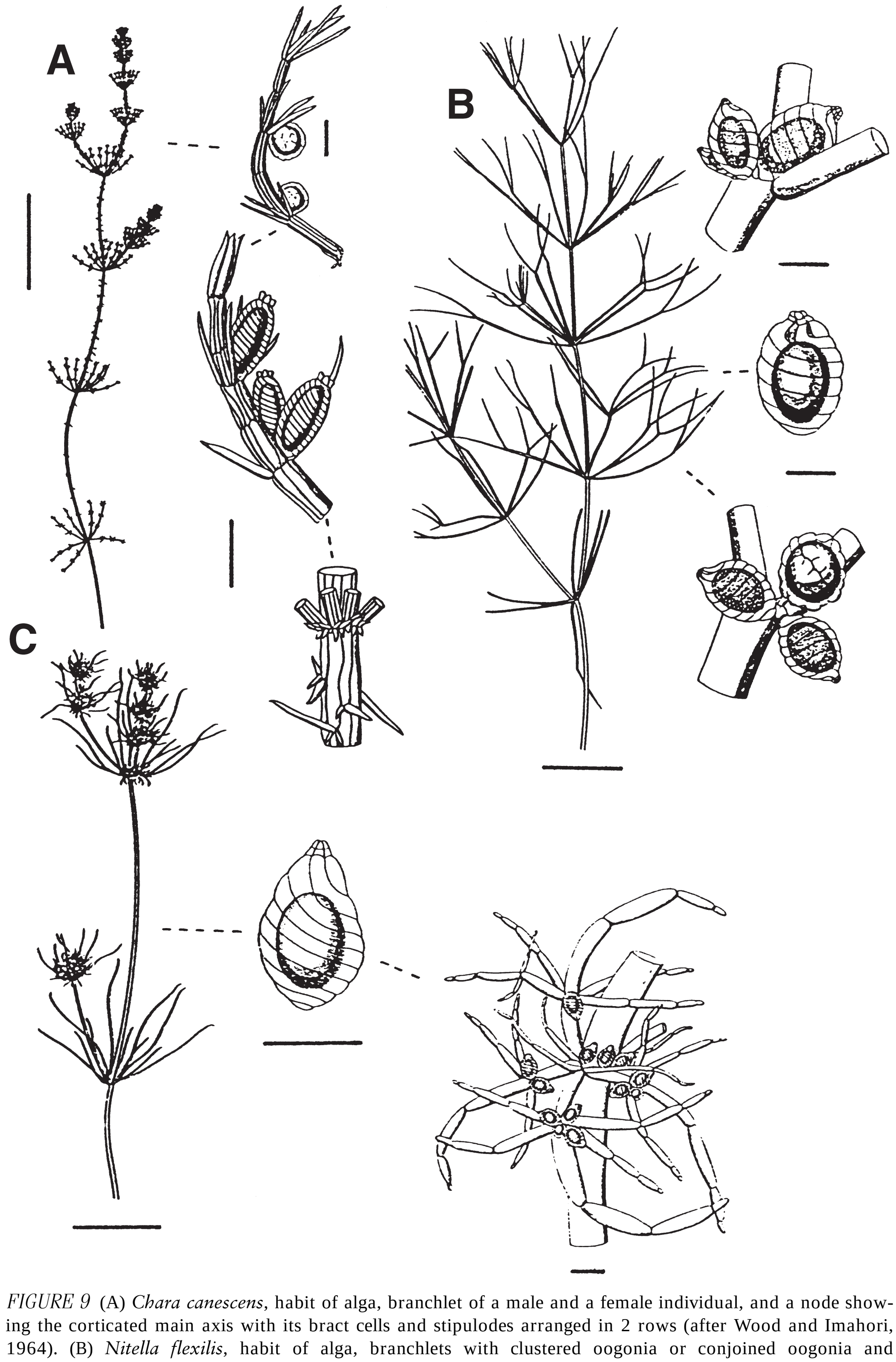

FIGURE 9 (A) *Chara canescens*, habit of alga, branchlet of a male and a female individual, and a node showing the corticated main axis with its bract cells and stipulodes arranged in 2 rows (after Wood and Imahori, 1964). (B) *Nitella flexilis*, habit of alga, branchlets with clustered oogonia or conjoined oogonia and antheridia, and an oogonium with a 2-tiered corona (after Wood and Imahori, 1964). (C) *Tolypella nidifica*, habit, portion of node with fertile branchlets, and an oogonium with a 2-tiered corona (Wood and Imahori, 1964).

Chamaetrichon Tupa (Fig. 1E)

Filaments are uniseriate, creeping, and simple or with a few side branches. Cells each contain a parietal chloroplast and a single pyrenoid. Asexual reproduction is by naked, symmetrical, quadriflagellate zoospores, each with a parietal chloroplast, pyrenoid, stigma, and two contractile vacuoles. Aplanospores are known.

This is a little-known genus, with the only traced record accompanying Tupa's description of the type (*C. capsulatum*) (Tupa, 1974). She isolated it from the surface of a submerged liverwort and the leaves of *Sagittaria* growing in a shallow drainage area of Double Lake, Sam Houston National Forest, near Coldspring, San Jacinto County, Texas.

It is structurally similar to the genus *Protoderma* but differs from it in producing quadriflagellate rather than biflagellate zoospores.

Chlorochytrium Cohn (Fig. 1N)

Chlorochytrium is endophytic and consists of solitary and spherical cells, irregularly curved or lobed, often with walls thick and lamellate. Cells each possess a chloroplast whose arms radiate from center towards the periphery where they expand into broad parietal lobes or a complete parietal layer, with one or several scattered pyrenoids. Reproduction is by biflagellate zoospores and gametes; aplanospores and akinetes known.

Chlorochytrium commonly develops in the intercellular spaces of various aquatic plants (especially the duckweed *Lemna*), or is associated with soil or other algae (including cyanobacteria); it is mostly within animal and plant tissues, only more rarely reported free-living. Five species have been reported from North America.

The taxonomic status of *Chlorochytrium* needs reevaluating since some species have been demonstrated to be life history stages of other filamentous algae (Chapman and Waters, 1992). Its exact taxonomic placement remains uncertain; often it has been regarded as belonging to the Chlorococcales. The genus is in need of revision.

Chlorotylium Kützing (Fig. 3F)

Chlorotylium forms hemispherical to irregularly cushion-shaped clumps, lime-encrusted, and consists of uniseriate and almost wholly unilaterally branched filaments. Cells are of two types: short cells often in pairs, each possessing a green-colored or often reddish parietal chloroplast and a pyrenoid, alternating with longer and sometimes colorless cells, with each containing a pyrenoid.

It is most commonly found in flowing water especially streams and areas of rapids. Of the five known species, at least two occur in North America.

Cloniophora Tiffany (Fig. 3G)

The filaments are uniseriate, erect and with downwardly growing rhizoidal branches, the secondary branches considerably smaller than the primary branches and irregularly arranged, with the terminal cell conical or its apex blunt. Cells are cylindrical, inflated, or capitate, each possessing a large parietal chloroplast and one to several pyrenoids. Zoospores are reported in one species, biflagellate, arising in cells of lateral branches.

Cloniophora is often attached to rock surfaces in usually fast-flowing water; it is also reported coastally from freshwater seepages well above the high tide. Islam (1961) suggests that it probably grows in habitats where the salt content is higher than usual fresh water. It is generally considered to be a subtropical or tropical genus, and is widely distributed in Central America and the Caribbean and most common along the eastern coast of the United States (Florida to New Brunswick).

Coleochaete Brébisson (Fig. 2I, J)

Filaments are uniseriate, dichotomously, or irregularly divided, consisting of only a prostrate system or a prostrate and an erect system, often the latter system is sometimes initially unbranched; prostrate forms have loose and spreading or laterally coalescing filaments forming a pseudoparenchymatous disklike expansion. Cells each contain a single, parietal, platelike chloroplast, one or two large pyrenoids, often bearing long, basally sheathed bristles. Zoospores biflagellate, one per cell. Sexual reproduction is oogamous: male gametes biflagellate, one per cell; oogonia are flask-shaped and bear a long trichogyne.

Coleochaete is frequently epiphytic (rarely endophytic) on submerged aquatic macrophytes, such as the submerged stems of cattail (*Typha*), and macroalgae, including the stonewort *Nitella*. It is also conspicuous on artificial surfaces including glass and crockery and frequently forms disklike colonies on the glass sides of aquaria (Prescott, 1983).

For a review of the genus, see Szymanska and Spalik (1993). Placed in the order Coleochaetales based on ultrastructural details.

Conochaete Klebahn (not pictured)

Conochaete grows as gelatinous cushions consisting of loose clusters of subglobose cells, frequently depressed, each cell bears a few long and delicate bristles arising apically on a mamillate wall protuberance or with an elongated gelatinous sheath. Cells each possess one or two parietal chloroplasts and a single pyrenoid, an oil droplet is often evident. Zoospores are usually four or eight per cell, the flagella number is unknown. Sexual reproduction is also unknown.

Only rarely reported, *Conochaete* is normally epiphytic on aquatic plants in upland areas. It is placed in the order Coleochaetales based on ultrastructural details.

Ctenocladus A. Borzi (Fig. 5B)

Ctenocladus is gelatinous and globular, consisting of erect uniseriate filaments, more or less radiating, with branches arising unilaterally. Cells are cylindrical and elongated, each containing a parietal chloroplast and one to several pyrenoids. Asexual reproduction is by biflagellate, micro- and macrozoospores arising in terminal cell or a series of cells. Quadriflagellate zoospores are reported in two undetermined species. Sexual reproduction is isogamous and gametes are biflagellate. Akinetes are spherical to subspherical, thick-walled, terminal, or in series.

Ctenocladus grows on rocks and higher plants (usually *Salicornia*) in inland saline environments, especially ponds and the marginal shallows of lakes. It is known mainly from more arid parts of western North America. According to Blinn and Stein (1970), its distribution pattern in North America suggests that migratory waterfowl play an important role in its spread. It often occurs where the Solonchak and Solonetz soil groups are present; these soils contain a high percentage of high exchangeable sodium. It is one of the more important food sources for brine flies in saline lakes (Herbst and Castenholz, 1994).

All the available evidence suggests that *Ctenocladus* consists of a single very polymorphic species, *C. circinnatus*.

Desmococcus F. Brand (Fig. 1C)

Desmococcus forms solitary or sarcinoid packets of cells (see also Chap. 7), occasionally as short uniseriate filaments. Cells are normally spherical, each possessing a parietal chloroplast and a single pyrenoid. Aplanosporangia are large, spherical, and the surface of each spore is warty or punctate.

According to Smith (1933), it is "the commonest green alga in the world," and often forms a green coating along with *Apatococcus* on soil-free and damp surfaces including the trunks of trees, stone or brick walls, and wooden fencing; it is often very abundant in both shaded and polluted habitats where lichens are less uncommon.

In the past, considerable taxonomic confusion has surrounded it and closely related genera (e.g., *Apatococcus*, *Chlorococcum*, *Pleurococcus*). It is retained in the Order Chaetophorales, but has been placed in the Ctenocladales.

Dicranochaete Hieronymus (Fig. 2C)

Dicranochaete is solitary, or more rarely, cells united in short rows consisting of spherical or kidney-shaped cells with the apical portion of each usually ornamented and forming an operculum, laterally bearing one or more repeatedly dichotomously divided bristles. Cells each containing a single chloroplast in the form of an inverted parietal cup, with or without a few pyrenoids. Zoospores biflagellate, four to 32 per sporangium. Gametes biflagellate and fuse to form a quadriflagellate zygote; the resting spore thick-walled.

Rare, reported as an epiphyte on filamentous algae and submerged aquatic plants in lakes.

Placed in the order Coleochaetales based on details of its ultrastructure.

Dilabifilum Tschermak-Woess (Fig. 3K)

The filaments are uniseriate, consisting of a creeping primary system bearing secondary or tertiary branches and erect ones. Cells are cylindrical in younger filaments, usually swelling to form akinetes in aging cultures; on agar it develops into dark green mass of coccoid cells. Cells each contain a parietal chloroplast and a pyrenoid. Zoospores quadriflagellate, produced in swollen intercalary cells. Aplanospores 16 or 32 per cell.

Isolated from various surfaces (e.g., lichens, mollusk shells, calcareous crusts, stones), it is collected in freshwater and marine (intertidal) habitats (more frequently recorded from the former). It is only known from Europe and North America.

Dilabifilum is a rather ill-defined genus in which reports of sporangia and quadriflagellate zoospores require further confirmation; asexual reproduction has yet to be discovered in two species.

Draparnaldia Bory de St.-Vincent (Figs. 3C, D)

A soft mucilaginous envelope encloses the erect uniseriate filaments of *Draparnaldia* that are attached basally by rhizoidal branches; the principal erect branches bear oppositely, alternately, or in whorls, lateral clusters or tufts of smaller-celled and much divided secondary-branches, each terminating in a blunt cell or multicellular hair. Cells of main branches are barrel-shaped or cylindrical, each containing a large parietal, band-shaped, entire or reticulate chloroplast with a smooth or laciniate margin, and several pyrenoids. Cells of secondary branches each possess a single, laminate chloroplast covering almost the entire wall, with one to three pyrenoids. Asexual reproduction is by biflagellate zoospores and thick-walled aplanospores. Sexual reproduction is isogamous by quadriflagellate gametes.

Draparnaldia is most frequently encountered in pristine upland streams and spring areas. It is more common and abundant in cooler waters, hence it is more likely to be encountered in the spring and autumn months in temperate areas. About 20 species are known, of which at least five are reported from North America.

The genus exhibits considerable morphological plasticity in response to differences in the environment, leading to considerable doubt concerning the validity of characters traditionally used for taxonomic discrimination. In nutrient-enriched water, species of *Stigeoclonium* are capable of displaying "draparnaldioid" features (Simons *et al*., 1986), but are unlikely to be mistaken for a *Draparnaldia* due to the absence in *Stigeoclonium* of distinctive clusters of secondary branches and the smaller diameter of main axes (< 30 μm).

Draparnaldiopsis G. M. Smith *et* Klyver (Fig. 3B)

Similar to *Draparnaldia*, but the principal branches of *Draparnaldiopsis* are rarely divided, consisting of alternating short and long cells; clusters of secondary branches arise opposite one another or in whorls and only from short cells, basal cells of secondary branches are usually di- or trichotomously branched, each branch generally terminates in narrowly obtuse cells or a long hair. Cells of secondary branches are cylindrical, inflated, or fusiform, each with a bandlike or reticulate chloroplast as in *Draparnaldia*. Zoospores are often produced in cells of main axes and gametes in cells of secondary branches. Zoospores are of different sizes: larger zoospores quadriflagellate (two flagella of different length), smaller ones biflagellate. Gametes are isogamous and biflagellate.

Only two species are known from North America, both from lakes in the western United States.

Entocladia Reinke (Fig. 1K)

The *Entocladia* filaments are uniseriate, creeping, and irregularly divided, sometimes coalescing to form a compact pseudoparenchymatous and one-celled layer prostrate expansion; branches are occasionally short and grow mostly by division of bluntly pointed terminal cells, with some species bearing from the upper cell wall a long, tapering bristle (sometimes basally swollen). Cells each contain a parietal, platelike chloroplast, with one or more pyrenoids. Zoospores quadriflagellate. Sexual reproduction is by biflagellate isogametes.

It is endo- or epiphytic on or within the wall layer of filamentous green algae such as *Cladophora*, *Rhizoclonium*, and *Pithophora*, or growing among the cells of the red alga *Lemanea*. Some species have been regarded as mere juvenile stages in the development of *Stigeoclonium*. Most species are marine rather than freshwater.

The exact taxonomic placement of freshwater representatives remains uncertain, as does the relationship of these to the marine species. The genus is in need of further investigation.

Fridaea W. Schmidle (Fig. 3E)

Fridaea forms hemispherical, compact, heavily lime-encrusted tufts, usually yellow-green in color, consisting of a prostrate and erect system of branched uniseriate filaments bearing tapering bristles as an extension of the distal cross wall of a cell, occasionally bristles separated from the cell by a wall. Cells are cylindrical and elongated, each possessing a very pale green, parietal chloroplast confined to the distal end, with one to four pyrenoids present. Cells interpreted as zoosporangia arise laterally, elongate and bottle-shaped, zoospores or swarmers unknown. Akinetes is uncertain.

Only known from moving water, it very rarely is recorded, but when present it is often abundant. The genus is monospecific.

Fritschiella Iyengar (Fig. 3H)

The filaments of *Fritschiella* are uniseriate, consisting of colorless downward-growing rhizoidal filaments and prostrate subterranean filaments, the latter often crowded and irregularly divided. Above ground the tufted, irregularly branched erect filaments have cells becoming longer and more elongated toward the apex, the terminal cell is conical. Cells above soil each contain a parietal chloroplast and several pyrenoids. Reproduction is confined to subterranean filaments. Zoospores are of three types: quadriflagellate macro- and microzoospores, and biflagellate microzoospores. Gametes are biflagellate and isogamous.

Fritschiella is terrestrial, growing more or less gregariously on moist soil or silt including drying rainwater puddles. It is known from the southwest United States.

Gomontia Bornet *et* Flahault (Fig. 1H)

The filaments of *Gomontia* are uniseriate, consisting of irregularly branched masses lying immediately below the surface of mollusk shells, wood, or limestone, sometimes densely crowded and pseudoparenchymatous, with downward-growing rhizoidal filaments. Cells are cylindrical, ovoid or polygonal, walls are thick and lamellate, each containing a dense laminate and sometimes netlike or lobed chloroplast. Several pyrenoids and nuclei are present. Zoosporangia and aplanosporangia are formed by enlargement of cells of short erect branches; the number of flagella is uncertain (variously reported to be two or four). Akinetes are produced.

Gomontia perforates the shells of mollusks, the carapace of turtles, limestone, submerged wood, or

"*Cladophora* balls." Most species are marine, with only two reported in North America from non-marine environments.

Gongrosira Kützing (Fig. 3J)

Commonly cushion-like, sometimes lime-encrusted, *Gongrosira* consists of a prostrate and erect system of uniseriate filaments, the prostrate system single to multilayered, loose or pseudoparenchymatous, giving rise to an erect system usually of short branches terminating in blunt tips. Cells are cylindrical or somewhat inflated, sometimes with thick and lamellate walls, each with a prominent parietal chloroplast and one to several pyrenoids. Zoosporangia are terminal or intercalary; zoospores are biflagellate. Aplanospores or akinetes are produced, the latter reddish in color.

It often forms a greenish layer or cushion-like mass on stones, gastropods, aquatic macrophytes (e.g., *Potamogeton*, *Myriophyllum*, *Najas*, *Chara*) and other hard surfaces, and is frequently most evident along the shallow margins of ponds, lakes, and rivers; it is only rarely recorded from soil.

Many species are of doubtful validity, with considerable overlap of morphological characters considered of taxonomic importance. Few species have been examined in laboratory culture (e.g., *G. debaryana*, *G. scourfieldia*), with one species (*G. pseudoprostrata*) defined only on culture-based characters. Some species are poorly known or doubtful, and others have been transferred to other chaetophoralean genera. It is retained here in the Chaetophorales, but sometimes is placed in the order Ctenocladales.

Hazenia H. C. Bold (Fig. 1I)

Hazenia has a tubelike sheath of soft mucilage containing irregularly branched uniseriate or partly multiseriate filaments; short lateral branches terminate in a conical cell. Cells are subquadrate to globose, not always in close contact, each contains a parietal chloroplast and a pyrenoid. Gametes are biflagellate and isogamous. Zoospores are unknown.

Only known from soil. *Hazenia* is monospecific.

Helicodictyon L. A. Whitford *et* G. J. Schumacher (Fig. 1M)

A spherical or irregularly shaped mucilaginous envelope surrounds the peculiarly curved or twisted and indistinctly branched filaments of *Helicodictyon*. Cells are short, curved, or irregularly lobed, each possess one to three short, parietal, platelike chloroplasts and a single pyrenoid. Vegetative cells become transformed into single biflagellate zoospores.

Helicodictyon is sometimes present in sufficient quantity in ponds (including fish hatcheries) and lakes to form a sulfur-yellow bloom, especially in autumn. It is most commonly reported in acid and slightly brown water along the coastal plain of the Atlantic seaboard of the United States, and is monospecific.

Older colonies are reported (Biebel, 1968; Whitford and Schumacher, 1969) to often have a large gas vacuole trapped within the center. It has been suggested that it may be more closely allied to the Order Chlorococcales than the Chaetophorales.

Leptosira A. Borzi (Fig. 1J)

Leptosira has cushion-like tufts consisting of erect and prostrate systems of uniseriate filaments, irregularly to subdichotomously branched, with side branches short and terminating in variously shaped cells. Cells are spherical, barrel-shaped, elliptical, or irregularly shaped, each possess a parietal chloroplast filling the lumen; starch is present, with pyrenoids not usually evident (sometimes masked by starch). Reproduction is by biflagellate zoospores and biflagellate isogametes. Aplanospores are spherical, four to several in each sporangium.

It grows epiphytically on plants such as mosses in springs and ponds, rarely abundant but widely distributed in North America. Only one species is commonly reported in North America of the five to six currently recognized.

Some uncertainty surrounds the validity of this genus. It very closely resembles *Pleurastrum* but differs from it by its indistinct pyrenoid (see the note under *Pleurastrum*). Wujek (1971) considered *Leptosira* to be identical to the monospecific genus *Leptosiropsis*, the latter originally separated from *Leptosira* by having an evident pyrenoid. A pyrenoid is normally present but is often only observed when examined with the electron microscope, or is sometimes obscured by a mass of starch (Lokhorst and Rongen, 1994).

Microthamnion Nägeli (Fig. 3L)

Microthamnion is microscopic, attached by a mucilage pad and consists of an erect system of uniseriate filaments, each side branch arise immediately below the cross wall, and its first cross wall is often remote from its point of origin, terminates in an obtuse cell. Cells are cylindrical, each contains a parietal chloroplast and no pyrenoid. Zoospores are biflagellate, anteriorly projecting outward to form a neck. There is doubt regarding sexual reproduction and formation of akinetes and aplanospores.

Microthamnion occurs on aquatic macrophytes, rotting wood, stones and other surfaces in the shallows of pools, slow-flowing streams, and ditches. It is also common in springs and grows on peat in bog pools. Sometimes it is entrapped in water film and is associated with soil or wet carpets of moss. Of those species present in organically polluted habitats (e.g., sewage treatment plant filters), it develops in such abundance as to form macroscopic growth. Only three species are currently recognized (see below), one known from North America.

The considerable morphological variation exhibited by the genus has given rise to taxonomic problems because of the unreliability of the trivial vegetative features traditionally used to define subgeneric taxa. Only three species are recognized by John and Johnson (1987), and of these, two are regarded as doubtful with *Microthamnion kuetzingianum* and *M. strictissimum* considered to be conspecific. Curved branches commonly develop on plants entrapped in the surface water film. It is traditionally placed in the Chaetophorales, but its correct position remains uncertain, although it is placed by Melkonian (1990) in its own order, the Microthamniales.

Oligochaetophora G. S. West (Fig. 2E)

Oligochaetophora has a mucilaginous envelope surrounding the loose clusters of two to six subglobose or ovoid cells. The walls are very thin and homogeneous, each cell bears two to four unbranched, spinelike bristles. Each cell has a single parietal chloroplast and two or three starch grains. Reproduction is unknown.

It is rare and epiphytic on submerged aquatic plants, monospecific, and is placed in the order Coleochaetales based on ultrastructure details.

Pleurastrum R. Chodat (Fig. 1F)

The cells are solitary or form sarcinoid packets each of a tetrad of cells, or as short-branched filaments. Cells are also spherical, each containing a parietal chloroplast (often lobed) and one or more pyrenoids. Reproduction is by zoospores, aplanospores, and akinetes. Zoospores are naked, biflagellate, variable in shape, and often asymmetrically flattened.

Pleurastrum has been collected and isolated into culture from standing water (often stagnant) and the surface of damp soil.

A chemotaxonomic investigation has concluded it to be impossible to separate certain taxa of *Pleurastrum* and *Leptosira*. These two genera are morphologically very similar and separation is based solely on the absence of a pyrenoid in *Leptosira*, a distinction that is no longer valid (see *Leptosira*). Ultrastructural investigation of *Pleurastrum* has revealed the presence of two types of pyrenoid structure: one type traversed by several membranes and the other by a single membrane. Seven species are currently recognized and separation is only considered possible if they are compared in culture.

Polychaetophora W. West *et* G. S. West (Fig. 2D)

Polychaetophora is solitary, or more usually, two to six subglobose, ellipsoid, or ovoid cells clustered together to form a pseudofilamentous mass, each cell with a very thick lamellate wall, sometimes with lamellate outgrowths, bearing eight to 12 delicate, flexuose, and unbranched bristles. Cells each have a single parietal chloroplast, often indistinct at the margins, with oil droplets and no pyrenoid(s).

Occurrence is rare, usually growing on aquatic macroalgae, and is monospecific. See remarks under *Oligochaetophora*. It is placed in the order Coleochaetales, based on ultrastructure details.

Protoderma Kützing (=*Ulvella* Crouan) (Fig. 1G)

The filaments of *Protoderma* are uniseriate, closely coalesce to form a pseudoparenchymatous, single-layered expansion with filaments free from one another towards the margin. Cells towards the center are usually polygonal and those towards the margin are more cylindrical, each possess a parietal platelike or flattened chloroplast, with or without a pyrenoid. Zoospores are biflagellate and generally form in central cells. Aplanospores are elliptical or spherical. The palmelloid stage is formed by repeated division of the central cells.

Protoderma is common in streams growing on rocks and the stems of submerged aquatic plants, including filamentous macroalgae, such as *Cladophora*.

It is almost impossible to separate *Protoderma* from other related genera, especially species in other genera in which the erect system is very reduced (e.g., *Pseudendoclonium prostratum*). Often separation is only possible if the flagellar number of the zoospores is known. Some authorities consider the genus to be monotypic with *Protoderma beesleyi* regarded as belonging to *Ulvella*, a largely marine genus to which it was originally attributed. The exact placement of the genus is problematic, with ultrastructural findings indicating that those species examined should be transferred to the order Ulvales. Until further species are examined, it is retained in the Chaetophorales, the order to which it has been traditionally assigned.

Pseudendoclonium Wille (Fig. 1L)

Filaments are uniseriate. An irregularly branched prostrate and erect system are present, with prostrate filaments loose and spreading or pseudoparenchymatous. Cells are cylindrical or inflated, each possessing a parietal chloroplast and a single pyrenoid. Zoospores are quadriflagellate. Aplanospores and zoospores often are produced in two- or four-celled sarcinoid packets of swollen cells, which are pleurococcoid in appearance. Akinetes are known.

Pseudendoclonium is isolated from stones and the surface of various submerged plants (e.g., *Ceratophyllum*, *Lemna*, *Vallisneria*, *Myriophyllum*, *Najas*, *Potamogeton*) and grows in a wide variety of water bodies including ponds, springs and slow-flowing rivers; it is rarely isolated from soil.

It is essential to know the flagellar number of the zoospores, otherwise it is impossible to separate it from species of *Gongrosira*. All the freshwater species have been described from isolates studied in laboratory culture. Square or diamond-shaped surface scales have been discovered on the zoospores of *P. basiliense* var. *brandii* and *P. akinetum*, but not on those of the marine species *P. submarinum*. Inconsistencies in ultrastructural features of the few species so far investigated inevitably cast doubt on whether the genus represents a natural grouping. Fine structural details of the flagellated stage have led to the opinion that it should be placed in the order Ulotrichales.

Pseudochaete W. West *et* G. S. West (Fig. 2I)

The filaments of *Pseudochaete* are uniseriate, prostrate, short, bearing from the sidewalls of some cells unbranched tapering filaments about five to eight cells in length. Cells are usually cylindrical, each possess a parietal chloroplast and a single pyrenoid only in cells of prostrate system. Zoosporogenesis and sexuality are unknown.

Only two species are known, both rare and often on submerged plants, logs, and on cattail stems.

Genus is incompletely known and doubtful. It has been suggested that the type species (*P. gracilis*) might be a developmental stage of *Stigeoclonium*.

Pseudulvella Wille (Fig. 1A)

Filaments are uniseriate, pseudoparenchymatous with radially arranged cells and several cells in thickness in the center of a disklike or somewhat irregular expansion, enveloped in mucilage and bearing a few hyaline bristles. Cells each possess a parietal chloroplast, netlike or divided into several oval bodies and a single pyrenoid. Zoospores are quadriflagellate, generally formed toward the center.

Only one freshwater species, *P. americana*, is reported on rare occasions from the United States (e.g., Michigan, Iowa).

Schizomeris Kützing (Fig. 7C)

Schizomeris has macroscopic filaments, unbranched, basally uniseriate, and becoming multiseriate above to form a solid parenchymatous cylinder with ringlike constrictions at intervals along its length. Cells are long and cylindrical toward the base, quadrate to angular above where multiseriate, thick-walled, each containing a parietal chloroplast encircling two-thirds of the cell, and toward the base cells possessing several pyrenoids. Zoospores are quadriflagellate, formed in multiseriate portions.

It grows as dark green, coarse clumps and is widespread but not common. Known from quiet water and in or near waterfalls. *Schizomeris* is often most common in fairly eutrophic water, including polluted environments such sewage sludge or shallow lakes close to discharge from sewage treatment works. Only two species are known.

The view that the genus represents a growth form of *Stigeoclonium* has not received general acceptance. On ultrastructural grounds, its affinities lie with the order Chaetophorales *sensu stricto*, although it is morphologically similar to members of the Ulvales.

Stigeoclonium Kützing (Fig. 3A)

Filaments are uniseriate with variously developed prostrate and erect systems: erect filaments alternately, oppositely or dichotomously branched, occasionally whorled or irregularly arranged, apices acute, narrowly obtuse, or each bearing a multicellular hyaline hair; prostrate filaments are creeping or rhizoidal, occasionally forming a pseudoparenchymatous disklike expansion. Cells are cylindrical or inflated, thick or thin-walled, each contain a single, parietal chloroplast, and one to several pyrenoids. Zoospores are quadriflagellate and of two sizes: micro- and macrozoospores. Gametes are isogametes and biflagellate or quadriflagellate.

Stigeoclonium often occurs as delicate tufts or mats on rocky surfaces and on submerged aquatic plants. It grows over a very wide habitat range, although is often especially common in swiftly flowing streams and rivers. It is sometimes very abundant in polluted rivers (e.g., immediately below sewage treatment works) including waters with relatively high levels of heavy metals where *Cladophora* may be absent.

This is an extremely polymorphic genus in which considerable doubt attaches to the validity of many described species. Species are best delineated only after growth in carefully defined culture conditions (Cox and Bold, 1973). Some modern culture-based studies (Simons *et al*., 1993) have recognized just three species and consider the majority of the 30 or more recognized species to be environmentally induced forms.

Thamniochaete F. Gay (Fig. 2B)

Filaments of *Thamniochaete* are uniseriate, creeping or erect branches are few and short, terminal and sometimes other cells also bear long and often basally inflated bristles or short spinelike projections. Cells are irregularly cylindrical, barrel-shaped, or inflated, each containing a parietal chloroplast and a single pyrenoid. One species produces ovoid or globose, biflagellate zoospores. Akinetes are known.

Thamniochaete grows as an epiphyte on larger algae or hard surfaces, usually in moving water. Only one of the three known species so far reported is from North America.

One species of *Endoclonium*, *E. rivulare*, is often placed in *Thamniochaete*.

Trichodiscus Welsford (Fig. 2G)

Trichodiscus is a disklike thallus reaching about 1 mm in diameter, consisting of branched uniseriate filaments; prostrate filaments radiate and centrally form a pseudoparenchymatous disklike expansion and are free at the margin, bear short erect branches and colorless (above the base) multicellular hairs arise from the center. Cells each have a lobed, parietal chloroplast and a single pyrenoid. Zoospores are biflagellate, produced in central cells of pseudoparenchymatous disklike expansion. Akinetes are produced by rounding up of cells.

This is a dubious genus, with its only species (*T. elegans*) discovered in the U.K. on the sides of a glass jar in which was grown an *Azolla* plant imported from North Carolina. It closely resembles some species of *Stigeoclonium* species (e.g., *S. farctum*) from which it is doubtfully distinguished by having somewhat inflated sporangia and gametangia. Due to the doubt attached to the genus and its putative North America record, it is not keyed out here.

Trichophilus Weber-van Bosse (Fig. 1B)

The filaments of *Trichophilus* are uniseriate and multiseriate, prostrate, irregularly divide to give short branches, partly pseudoparenchymatous. Cells are thick-walled, each contain a parietal chloroplast and a difficult-to-discern pyrenoid. Zoospores quadriflagellate and of two sizes.

It grows on or within the hair scales of sloths and on the shells of freshwater mollusks. Only three species are known.

Trichophilus closely resembles the genus *Entocladia* Reinke.

Order Charales

Order Charales is macroscopic, consisting of algae with creeping rhizoidal branches from which arise erect branches of limited growth, each bearing whorls of secondary branches (branchlets) of limited growth. It has complex and unique type of advanced oogamous reproduction. Only three of the seven genera are known from North America. The genus *Nitellopsis* is so far only reported from South America.

Chara L. (Fig. 9A)

Chara is macroscopic, usually lime-encrusted, with similar organization to other members of the order. Branches of a unlimited growth composed of elongated, single-celled internodes and multicellular nodes, with branchlets (rarely branches of unlimited growth) arising from the short cells of nodes; the cortex is single-layered over internodal cells, rarely partially or ecorticate, showing varying degrees of development of primary, secondary, and tertiary cell rows, with the primary row always distinguished by the presence of spine cells. Branchlets are undivided, subtended by a single or double ring of unicellular outgrowths (stipulodes); simplified cortication is over lower internodal cells; rings of unicellular bract-cells develop at nodes; the terminal segment is single-celled or in chains not separated by nodal cells. Reproduction is by an advanced form of oogamy, the oogonium always above the antheridium when together on a branchlet (monoecious species). Oosporangium each consists of an oogonium surrounded by eight spirally twisted sterile cells, bearing a crown of five cells. Antheridia are usually spherical consisting of eight shield-shaped cells borne on the end of a short stalk.

Thalli are often large (up to 1 m) and coarse, especially when heavily lime-encrusted. It is most abundant in hard water or alkaline ponds and lakes (also in the Great Lakes), where it can form extensive underwater meadows. It is found occasionally in the shallows of slow-flowing rivers and in spring seepage areas and known to grow to depths as great as 12 m (see also Chapter 2, Section II.F-2).

Often *Chara* species have a strong odor, hence its common name in North America of skunkweed or muskweed.

Lamprothamnium J.Groves (not pictured)

Lamprothamnium is macroscopic, with or without lime-encrustation and similar organization to other members of the order. Branches of limited growth are ecorticated. Branchlets are undivided, subtended by a single whorl of unicellular outgrowths (stipulodes), each acuminate and downwardly pointing, with rings of more or less equal unicellular bract-cells developing at the nodes. Oogonia arise below the antheridia at the branchlet nodes, each bears a crown of five cells in the form of a single ring, not compressed.

It is often large (reaching to c. 50 cm) and most common in coastal lakes and lagoons where the water is brackish. Considerable doubt attaches to *L. longifolium* and its variety *buckellii*, the only *Lamprothamnion* known from North America. It differs from all other members of the genus in being dioecious and having rudimentary cortical and spine cells. It is reported from a few widely scattered locations including British Columbia in Canada and Kansas in the United States.

Nitella C. Agardh (Fig. 9B)

Nitella is macroscopic, only lightly lime-encrusted or not at all; it is similar in organization to other members of the order but not as erect as *Chara*. Branches of limited growth ecorticate, hence spines are absent. Branchlets are not subtended by stipulodes and bract-cells are absent, forked one or more times into similar single-celled rays and/or one- to three-celled ultimate rays (sometimes divided two, three, or four times). Oogonia are produced at the forking of branchlets and occasionally at the base of a whorl of branchlets, with oogonia arising laterally and two to three together and antheridia solitary and terminal. Oogonia are with a crown of 10 cells in two tiers and laterally compressed

It is widely distributed and usually more common than *Chara* in softer water areas and acid lakes, including bog lakes where the water is stained brown. *N. flexilis* is common in streams throughout temperate regions of North America (Sheath and Cole, 1992).

Tolypella (A. Braun) A. Braun (Fig. 9C)

Macroscopic and not lime-encrusted, *Tolypella* is similar in structure to other charophytes, although less symmetrical, with dense heads and whorls of short branches and longer more disorganized branches. Branches ecorticate and hence are without spines. Branchlets have stipulodes or bract cells and are divided into unequal and multicellular rays. Oogonia and antheridia are in groups, often on the lower nodes of the branchlets or at their base, the oogonia having a corona of 10 cells in two tiers and not compressed.

Often somewhat scraggy looking plants. *Tolypella* often grows solitary in pools, ponds, ditches, shallows of hardwater lakes (including the Great Lakes), and slow-flowing streams. It is widely distributed in North America and occurs as far north as Newfoundland.

Order Cladophorales

Filaments are uniseriate, simple or branched, rarely forming a prostrate disklike expansion, more commonly erect and attached by rhizoids. Cells are quadrate to cylindrical, sometimes swollen, multinucleate, each containing a parietal and netlike chloroplast (more rarely numerous disklike chloroplasts), with numerous pyrenoids.

Basicladia Hoffman *et* Tilden (Fig. 5F)

Basicladia has a prostrate rhizoidal filament giving rise to a coarse erect filament only sparingly branched in the basal region. Cells are cylindrical toward the base and several times longer than broad, becoming gradually broader, shorter, and often barrel shaped toward the apex, walls are thick and lamellated, each possessing a parietal netlike chloroplast and numerous pyrenoids. Zoospores are biflagellate and produced in upper cells.

It is commonly epizoic on snails and on the upper carapace of several turtle species (occasionally on the head and tail), especially snapping turtles with the expression ‘mossback’ used for those animals with a thick algal growth. It is capable of growth on other surfaces provided they are hard and rough. Only four species comprise the genus with the two recorded from North America considered less common in northern parts of the United States than in the southern. For a review of its ecology, see Hutchinson (1975: 562-564) and Colt *et al*. (1995).

Chaetomorpha Kützing

Chaetomorpha is macroscopic, consisting of unbranched filaments, growth is intercalary, free living or attached basally by rhizoids. Cells are usually cylindrical to barrel shaped, walls lamellate, each with a parietal, netlike chloroplast and numerous pyrenoids and nuclei. Zoospores are bi- or quadriflagellate.

This is typically a marine genus with considerable doubt attaching to the report of it from freshwater in North America. The only report is that of *Chaetomorpha heningsii* growing in a roadside ditch in Washburn County, Wisconsin (Faridi, 1962).

Cladophora Kützing (Fig. 5C)

Macroscopic, free-living or attached by a disklike holdfast and/or downward growing rhizoids, consisting of erect or prostrate systems of filaments, sparsely to profusely branched, *Cladophora* is typically dichotomously or sub-dichotomously branched, with branches originating immediately below cross wall; growth is apical and/or intercalary. Cells are cylindrical to barrel shaped, each with a parietal and netlike chloroplasts, many nuclei and numerous, often bilenticular pyrenoids. Zoospores are bi- or quadriflagellate. Gametes are biflagellate and isogamous. Akinetes are rarely produced.

It is principally a marine genus, yet *Cladophora glomerata* is one of the most widely distributed and abundant macroalgae in fresh waters world-wide. It exhibits considerable morphological variation depending on environmental conditions, thus making identification difficult with many species records of doubtful validity. It is commonly attached but on occasion free-floating or lying as loose masses over sediments, frequently as long streamers attached to rocks in slow-flowing streams, and often forming entangled growths along lake shores; it is especially abundant in eutrophic habitats (e.g., parts of the Great Lakes) providing there is relatively low heavy metal contamination. One species is known to occur to a depth of 50 m in Lake Ontario. In a few North American lakes (e.g., in Massachusetts, New York, Indiana), it forms spherical to subspherical growths (ca. 2–10 cm in diameter) known as '*Cladophora* balls' (Daily, 1952; Kindle, 1934). It is also a nuisance alga commonly known as 'blanket weed' or 'cotton-mat' alga (see Chapter 24); the term might equally apply to other mat-forming green algae (e.g., *Oedogonium*, *Spirogyra*).

Considerable uncertainty surrounds the separation of many species whose identification often depends on combinations of overlapping character states. Culture studies have demonstrated that many earlier recognized species are merely growth forms or environmental variants (Hoek, 1982). Due to its world-wide importance, it has been the subject of several reviews (e.g., Whitton, 1970; Dodds and Gudder, 1992).

Dermatophyton A. Peter (Fig. 5D)

Epizoic, disklike, consisting of radiating rows of cells, *Dermatophyton* is polystromatic in its center and single-layered and dichotomously divided toward the margin. Cells are thick-walled, quadrate or subquadrate in the center, and more elongated close to the margin, each contain several nuclei, a parietal chloroplast and one to several pyrenoids. Zoospores are biflagellate, produced from flask-shaped cells in the center.

Thalli form deep green, irregular crusts reaching 5 mm across, growing on and penetrating cracks into the carapace of freshwater turtles where it develops between the layers of the lamellae. It grows on various turtles and sometimes accompanies the green alga *Basicladia*. It is monospecific.

It was transferred from Chaetophorales, where it was often placed due to presence of several nuclei in each cell.

Pithophora Wittrock (Fig. 5E)

Pithophora is macroscopic, consisting of freely branched filaments, lateral or occasionally oppositely branched, often side branches originate a short distance below the cross wall; occasionally unicellular or multicellular rhizoidal branches develop from some branch tips or base of filaments. Cells are cylindrical, often thick-walled, each with a parietal and netlike chloroplast as well as several pyrenoids and nuclei. Akinetes are thick-walled, barrel-shaped, or oval, terminal or intercalary in position.

It is known from rapids in rivers but is more common in ponds and lakes where it forms dense free-floating mats in shallow littoral areas; it is similar to *Cladophora* in that its prolific growth leads to lake management problems. Often somewhat coarser in texture than *Cladophora*, for this reason *Pithophora* is commonly known as the 'horsehair' alga. It is considered to be a tropical or subtropical genus but is widely distributed in temperate regions, especially in the eastern United States. Eight species are known from the United States alone.

Rhizoclonium Kützing (Fig. 5A)

Rhizoclonium consist of uniseriate unbranched filaments, occasionally short, unicellular, or few-celled rhizoidal branches arise at right angles to axis. Cells are usually more than twice as long as broad, sometimes with walls thick and lamellate, each with a parietal, reticulate chloroplast and several pyrenoids and nuclei. Reproduction is by fragmentation or akinetes. Zoospores biflagellate and gametes quadriflagellate.

It intermingles with other filamentous green algae usually as tangled mats in quiet stagnant water or as ropelike strands in flowing water. *Rhizoclonium hieroglyphicum* is the most common of the five freshwater species recorded from North America.

Order Cylindrocapsales

Filaments are uniseriate and unbranched, often becoming biseriate or multiseriate with age. See further details under *Cylindrocapsa*.

Cylindrocapsa Reinsch (Fig. 6I)

Filaments are unbranched, initially uniseriate, often becoming multiseriate or forming irregular, pseudoparenchymatous masses of cells. Cells are cylindrical, elliptical, rounded, oval, or subspherical, sometimes arranged in pairs, surrounded by a thick, rough, lamellated mucilaginous sheath, each containing an axial or stellate chloroplast in young filaments, sometimes chloroplast parietal, filling cell and often obscured by starch, with a single and often indistinct pyrenoid. Asexual reproduction is by biflagellate zoospores and aplanospores. Sexual reproduction is oogamous and monoecious: oogonia are greatly swollen and spherical or oval, dark green, with a thick lamellated wall; antheridia are in a series, two to four in each enlarged vegetative cell, producing two spindle-shaped biflagellate gametes. The zygote is spherical, thick-walled, and bright reddish when mature.

It is often attached but commonly become free floating and frequently intermingled with other filamentous forms growing in soft water lakes and bogs.

It is often assigned to its own order (occasionally to the Ulotrichales) although regarded by Hoek *et al*.

(1995) to be a filamentous representative of the Chlorococcales based on its type of mitosis and cellular division.

Order Microsporales

Filaments are uniseriate and unbranched, with separation of cells taking place at the point of weakness in the mid region of the wall, often resulting in the formation of 'H-shaped' portions on dissociation of filaments. If the filaments are correctly interpreted as being a row of autospores, then it is closely related to the Chlorococcales. According to Lokhorst (1999), ultrastructural details of the flagella apparatus provide yet further evidence that *Microspora* belongs to this order.

Microspora Thuret (Fig. 4E)

Free-living or basally attached at least initially, *Microspora* has a bipartite wall structure usually evident. Cells are quadrate, cylindrical, or slightly swollen, sometimes constricted at the cross wall, walls are often lamellate, each with a parietal chloroplast and perforated to produce a fine or coarse net, without pyrenoids. Reproduction is by biflagellate zoospores and biflagellate isogametes. Aplanospores and akinetes produced.

It is widely distributed in aquatic habitats, particularly among other filamentous algae in pools, streams, and ditches, especially during cooler times of the year. Some species are frequent in low pH environments, including bogs, marshes, and acid-mine drainage water.

Many characters traditionally considered of taxonomic importance are unreliable and are known to exhibit considerable variability within the same species depending on environmental conditions. Lokhorst (1999) has attempted to develop sounder species concepts by testing in laboratory culture the reliability of those morphological characters traditionally used to define species. It is possible to confuse *Microspora* with the Tribophyte (xanthophyte) *Tribonema* (see chapter 11), which is readily distinguished by having two or more, pale green or golden color, disklike chloroplasts and no starch.

Order Oedogoniales

Filaments are uniseriate, branched or unbranched, with or without bristles, often characterized by having one or more ringlike thickenings ('cap cells') at the distal end of some cells. It has a complex and unique form of oogamous reproduction.

The three genera represent some of the most comprehensively researched in North America thanks to the classic work of Tiffany (1930) on the Oedogoniales of the United States.

Bulbochaete C. Agardh (Fig. 8A)

The filaments of *Bulbochaete* are unilaterally branched, usually attached, consisting of cells broader at the upper ends and the majority bearing a long, basally swollen, colorless bristle. Cells are broader at the upper end, each containing a parietal, netlike chloroplast, with several pyrenoids. Zoospores have an apical ring of flagella, one produced per cell. Sexual reproduction is oogamous: large swollen female oogonia, boxlike antheridia, or dwarf male plant (short filaments) grow on or near the oogonia; the zygote is thick-walled and sometimes distinctively ornamented.

It is commonly epiphytic on submerged plants, submerged portions of overhanging and larger filamentous algae in slow-flowing water and pools and ponds. Of almost 60 species known, over half occur in the United States.

Oedocladium Stahl (Fig. 8B)

The filaments of *Oedocladium* are profusely branched, consisting of prostrate and erect portions; rhizoidal branches are produced in terrestrial and aquatic environments with these colorless in former, often with the apical cell conically pointed and having a distinct cap. Cells are cylindrical, each with a parietal and netlike chloroplast, with several pyrenoids. Reproduction is similar to that of *Oedogonium*. Akinetes usually developing on rhizoidal branches, solitary or in short chains and reddish in color.

It is most common along moist stream banks and on sandy loam, often growing together with moss protonema, thalloid liverworts, *Vaucheria*, and other soil algae.

Oedogonium Link (Fig. 8C)

Oedogonium filaments are unbranched, usually attached and without bristles. Cells are cylindrical, sometimes slightly broader at the anterior end, characterized by one or more ringlike caps immediately below the cross wall, each containing a parietal, netlike chloroplast and several pyrenoids. Zoospores have a distinctive apical ring of many flagella, one produced per cell. Sexual reproduction is oogamous: large swollen female oogonia, boxlike antheridia or some species with dwarf male plants (short filaments) grow on or close to the oogonia; zygote is thick-walled and sometimes distinctively ornamented.

It is free-living and sometimes epiphytic on submerged plants growing in similar aquatic habitats to *Bulbochaete*. Over half of the more than 250 known species are recorded from the United States.

Species-level identification is very difficult since they are based mainly on reproductive characters. Only rarely is the alga discovered in the reproductive condition.

Order Prasiolales (=Schizogoniales)

Filaments are uniseriate or multiseriate and form solid cylinders or expanded sheets of cells. See the description of *Prasiola* for further details.

Prasiola C. Agardh (Fig. 7E)

Prasiola is terrestrial or aquatic, growing as two morphological forms: single-layered (rarely two-layered), lobed, or ruffled and membrane-like, or as unbranched filaments (*Rosenvingiella*-stage). Flat expansions consist of quadrate or polygonal cells arranged in square to rectangular blocks, sometimes in smaller groups of four, often shortly stalked and attached by hairlike rhizoids. The filamentous stage is initially uniseriate, often becoming multiseriate, attached by rhizoids. Cells each contain a large, stellate chloroplast (both forms) and a single central pyrenoid. Reproduction is by nonmotile aplanospores produced in areas of thickening in membrane-like forms. Sexual reproduction is by oogamy: reproductive tissues (haploid) develops in the upper part of membrane-like stage, some areas producing biflagellate male gametes and others nonmotile egg cells.

Prasiola occurs in wide range of habitats that include terrestrial, marine, and freshwater. Some species grow on moist soil, rocks, stones, old walls, and trunks of trees, and others in cold, swift-flowing mountain streams. This genus is abundant in terrestrial habitats rich in nitrogeneous compounds, and hence its frequent association with the guano in bird roosts, as well as the base of urban walls and trees where animals frequently urinate. Other species (e.g., *P. mexicana*) colonize rocks in clear, swiftly flowing streams (Wehr and Stein, 1985; Sheath and Cole, 1992). The genus occurs over a wide geographical range from Arctic Canada in the north of North America to the tropics.

Considerable uncertainty surrounds the relationship between *Prasiola* and *Rosenvingiella* with some authors (e.g., Edwards, 1975) considering the latter to be a growth form of *Prasiola*. Rindi *et al*. (1999) observed a *Rosenvingiella* stage of *Prasiola* in Galway, Ireland and also recognize as distinct the genus *Rosenvingiella*; molecular studies are needed to resolve whether the latter is a separate genus. In the current treatment, it has been decided to regard *Prasiola* and *Rosenvingiella* as separate genera, but to recognize that the former has a *Rosenvingiella* stage. Uncertainty surrounds the exact placement of the genus in the classification of the green algae, although *Prasiola* and *Rosenvingiella* form a well-defined clade in molecular trees (Sherwood *et al*., 2000).

Rosenvingiella P. C. Silva

This is a terrestrial alga growing as unbranched, uniseriate, or multiseriate filaments attached at intervals by rhizoids, or parenchymatous, cylindrical and sometimes constricted, with cells sometimes in four-celled groups; chloroplasts are stellate and possess a single central pyrenoid. Reproduction is by nonmotile asexual spores.

It grows in many of the same habitats as *Prasiola*.

Considerable uncertainty surrounds the relationship between *Prasiola* and *Rosenvingiella* (see comments under *Prasiola)*.

Order Sphaeropleales

Filaments are uniseriate, unbranched, consisting of exceedingly long cells whose cytoplasm is divided by numerous large vacuoles into separate cytoplasmic units. Cytoplasmic units each are multinucleate and contain several a band-shaped or disklike chloroplasts.

Sphaeroplea C. Agardh (Fig. 6E)

Sphaeroplea cells are 2 to 60(-90) times longer than broad, divided into several large central cavities by a succession of transverse units of cytoplasm, each multinucleate unit containing several narrow, bandlike chloroplasts (occasionally perforate with age) and many pyrenoids. Reproduction is by fragmentation and production of biflagellate zoospores. Sexual reproduction is oogamous and vegetative cells do not change shape on becoming reproductive cells, monoecious, or dioecious. Oogonia are spherical, often lying in a single or double series, each having a conspicuous receptive spot; antheridia produces up to 300, spindle-shaped, reddish, biflagellate antherozoids. Zygotes have thick and ornamented walls, bright red when mature.

Free-floating and of sporadic occurrence, it is specially common in seasonally flooded grasslands and areas of gravel. If *Sphaeroplea* is present in considerable abundance, it causes the discoloration of the water on producing its red-colored zygotes. Only two species are reported from the United States.

Species are distinguished mainly upon features associated with zygote shape and ornamentation of its thickened wall (Ramanathan, 1964).

Order Trentepohliales

The filaments are uniseriate, branched, sometimes coalescing to form a pseudoparenchymatous and discoid expansion. Cells are each uninucleate, often red or reddish-orange due to an accumulation of haematochrome or astaxanthin, with a reticulate chloroplast or several discoid chloroplasts and without pyrenoids.

Cephaleuros Kunze (Fig. 6A)

Cephaleuros is epiphytic and endophytic on leaves, stems, and fruits of terrestrial plants. It consists of regularly branched coalescing filaments forming a prostrate and a one to several-layered pseudoparenchymatous mass from which downward-growing, irregularly branched filaments penetrate the deeper tissues and upward-growing, unbranched filaments terminate in a hair or group of sporangia. Cells are cylindrical, regularly or irregularly shaped, each possess a parietal and a netlike chloroplast or several discoid chloroplasts, without pyrenoids. Asexual reproduction is by zoosporangia produced in clusters terminally on erect hairs, zoospores biflagellate. Sexual reproduction is by enlargement of cells in the pseudoparenchymatous mass and produce biflagellate gametes.

It is considered a parasite or semi-parasite of terrestrial plants *(Magnolia*, *Rhododendron*, banana, tea, etc.) in the tropics and subtropics where it occurs on the surface, beneath the cuticle, and/or within the intercellular spaces and leads to the discoloration and degeneration of the tissues of the host. Often the parasitized areas are gray-green or reddish-orange in color and on the surface it forms cushion-like growths and has a velvety appearance. It is known to occur on many host species with Holcomb (1986) recording it from over 200 plant taxa in and near Louisiana. A serious parasite on some plants, it results in economic losses (especially to tea growers), but infections are usually easily controlled. For further information, see Chapman and Waters (1992) and Thompson and Wujek (1997).

Phycopeltis Millardet (Fig. 6B)

Phycopeltis is epiphytic on leaves of terrestrial plants. It consists of regularly or irregularly branched, coalescing filaments forming a prostrate, single-layered, pseudoparenchymatous discoid, lobed or irregularly shaped crust. Cells are cylindrical, each possess a parietal, netlike chloroplast or many irregularly shaped and platelike chloroplasts, often orange or yellowish due to haematochrome pigment, with a single pyrenoid. Asexual reproduction is by quadriflagellate zoospores, produced in sporangia arising on curved, one to multicellular stalks. Sexual reproduction is by biflagellate isogametes formed in terminal or intercalary gametangia.

It forms yellowish-green to deep orange colored expansions on leaves, most common in the humid tropical and subtropical regions where it causes commercial losses to cultivated plants.

Printzina Thompson *et* Wujek (not pictured)

Printzina is epiphytic. It consists of a widely spreading network of prostrate, branched filaments bearing few to many erect branches if present. Cells are cylindrical, barrel-shaped, globular, or elliptical, with numerous discoid chloroplasts, green color usually obscured by haematochrome, with pyrenoids absent. Reproduction is similar to that in *Trentepohlia* but differs from it by having globular to kidney-shaped rather than ovoid sporangia, the papilla pore basal on the sporangium rather than apical, sporangia lateral and always solitary and never grouped, and gametangia lateral or terminal rather than terminal.

It grows on the leaves and bark of trees growing in deep shade and high humidity and is often green in color.

The type species of the genus (*P. ampla*) is known from Columbia and Costa Rica where it grows on trees belonging to the families Melastomaceae, Mimosaceae, and Monimiaceae. In creating the genus, Thompson and Wujek (1992) transferred to it nine *Trentepohlia* species.

Stomatochroon Palm (Fig. 6D)

Subaerial and epiphytic, *Stomatochroon* consists of a lobed basal cell within a stomatal chamber, from the basal cell arises a cylindrical and erect, few-celled branch or hair bearing one or more sporangia. The basal cell contains chloroplasts in the lower part and the erect cell is brick red in color due to presence of carotenoid pigments.

It usually grows on the leaves of broad-leaved evergreen plants in humid tropical regions where it causes wounding responses and local damage when its growth is prolific (Chapman and Waters, 1992; Thompson and Wujek, 1997).

Trentepohlia Martius (Fig. 6C)

Trentepohlia is subaerial. It consists of variously developed prostrate and erect systems of sparingly to profusely branched filaments, irregular, alternate, or rarely oppositely divided, sometimes erect system very reduced, often terminal branches gradually attenuated and apical cells sometimes bearing a cellulose cap. Unicellular and cylindrical hairs are sometimes present. Cells are cylindrical, ellipsoidal, or beadlike, rarely more than twice as long as broad, walls are thick and lamellate, each with numerous, discoid or band-shaped chloroplast; chlorophyll usually is obscured by red or orange carotenoid pigments; pyrenoids are absent. Asexual reproduction is by quadriflagellate zoospores produced in terminal sporangia, or of two types of stalked lateral sporangia: ‘funnel sporangia’ consists of an apical portion of a terminal cell and a septum lying between two superimposed ringlike thickenings, and a sporangium with a bent stalk. Sexual reproduction is by biflagellate gametes produced in lateral or terminal, spherical gametangia. Akinetes are thick-walled and generally in series.

It forms greenish, yellowish-orange, or orange feltlike patches on leaves, twigs, stone walls, wooden fencing, and poles, and is often very extensive on rocky cliffs and moist soil in ravines and the damper sides of tree trunks. It is the second most common algal component of lichens where almost impossible to identify to species since it is frequently only present as single cells or short filaments without reproductive organs.

The genus *Physolinum* is now usually considered a synonym of *Trentepohlia*.

Order Siphonales

Filaments are tubular, with a single multinucleate cell, rarely divided by transverse walls, containing numerous discoid chloroplasts; pyrenoids are present or absent.

Dichotomosiphon Ernst (Fig. 6F)

Filaments of *Dichotomosiphon* are coenocytic and tubular, repeatedly dichotomously divided and transverse constrictions at the base of each dichotomy and additional ones often between successive branch divisions. Branches possess numerous discoid or ellipsoidal chloroplasts, often absent from rhizoidal branches at plant base; pyrenoids are absent. Asexual reproduction is by large akinetes produced terminally on lateral rhizoidal branches. It is oogamous, with gametangia borne on di-, tri- or tetrachotomously divided branches: oogonia are very large, yellowish, spherical, and arise on a curved supporting branch, antheridia are the same diameter as the supporting branch, curved and borne immediately below oogonia. Zygotes possess a smooth thick wall.

It grows as dense, tangled tufts or mats on streams and lake beds where it may occur in depths as great as 20 m; it often becomes buried in soft sediment (always sterile in deep water).

Phyllosiphon Kühn (Fig. 6H)

Phyllosiphon is endophytic on various members of the higher plant Family Araceae. It consists of coenocytic tubular branches (multinucleate and noncellular), profusely dichotomously or irregularly divided and interwoven with one another. Branches contain numerous discoid or elliptical chloroplasts except at apices; pyrenoids are absent. Reproduction is by many small ellipsoidal aplanospores.

It forms green patches within leaves and stems and often induces discoloration of the area of host tissue infected. Known from northern and eastern parts of the United States growing within the tissues of *Arisaema triphyllum*, it is commonly known as Jack-in-the-Pulpit (Smith, 1933, 1950). For further information, see Chapman and Waters (1992).

Protosiphon Klebs (Fig. 6G)

Protosiphon is a soil alga having above ground a swollen sacklike vesicle and an elongate, colorless, rhizoid-like portion extending downward into the soil. Cells are multinucleate, possessing a netlike chloroplast and several pyrenoids. Reproduction is by many biflagellate zoospores and isogametes produced within the vesicle. Cytoplasm sometimes divides in multinucleate portions and these develop into a thick-walled resting stage (known as coenocysts).

It is considered to be fairly common and widespread on moist soil where it often forms conspicuous green or orange patches.

Order Ulotrichales

Filaments are uniseriate, rarely multiseriate, unbranched, sometimes exhibiting basal-apical polarity when attached. Cells usually are quadrate or cylindrical, uninucleate, each possess a single, parietal, laminate, or broadly discoid chloroplast, bandlike or more usually only partially encircling cell, with or without pyrenoids.

This is a very heterogeneous order, with several genera now placed in other orders based on characters associated with mitosis and cellular division as well molecular data (e.g., *Uronem*a, *Klebsormidium*; see comments below). The affinities or many genera remain unclear and, therefore, most are retained in this order.

Binuclearia Wittrock (Fig. 4B)

The filaments consist of long cells usually containing pairs of oval or subspherical protoplasts. There are spaces between protoplasts and the cross wall have transverse lamellations of a gelatinous material, each protoplast with a single laminate chloroplast, incompletely or completely encircling cell, usually with pyrenoid (rarely absent).

It often occurs intermingled with other filamentous algae and submerged aquatic macrophytes, especially in softwater ponds and bogs.

Elakatothrix Wille (Fig. 4Q,R)

Elakatothrix is solitary or colonial, with cells contiguous and arranged in pseudofilaments embedded in a spindle-shaped and unstructured mucilaginous sheath, with the sheath margin definite or somewhat irregular in outline. Cells are spindle-shaped, cylindrical, oval, or elliptical, apices blunt or pointed, walls thin and hyaline, each contain one or three parietal, laminate, cup- or girdle-shaped, straight or spiral chloroplasts, and one or two pyrenoids. Asexual reproduction is by aplanospores. Sexual reproduction is unknown.

It is a widely distributed planktonic alga, although its juveniles are sometimes epiphytic (Lokhorst, 1991). Of the five known species, only one is known only from the United States, namely *Elakatothrix americana*.

It is often now placed in the Order Klebsormidiales based on ultrastructural features.

Filoprotococcus (H. W. Nichols *et* H. C. Bold) Thompson *et* Wujek (Fig. 7B)

Filaments are initially uniseriate, soon becoming multiseriate and transformed into a series of sarcinoid packets of cells which eventually dissociate. Cells each contain a parietal chloroplast and a single pyrenoid. Zoospores are quadriflagellate, produced singly or two to four in each cell of multiseriate filaments and sarcinoid packets. Sexual reproduction is unknown.

It has been isolated from cultures of soil from Texas and *Tolypella* collected from a marsh near Lawrence, Kansas (Thompson and Wujek, 1996). It is probably more widely distributed than records indicate.

Thompson and Wujek (1996) transferred *Trichosarcina polymorphum* to this genus although recognizing minor differences between the two (e.g., two to four zoospores per cell in the plants examined by them from Kansas and only one in those described from soil cultures).

Geminella Turpin (Figs. 4H, I)

Geminella filaments consist of cells separated but in a loose linear arrangement and equidistant, lying in pairs or contiguous, surrounded by a thick, cylindrical mucilaginous sheath. Cells are generally longer than broad, cylindrical with rounded apices, inflated and ellipsoidal, oval or barrel-shaped, each contain a parietal, girdle-like or laminate chloroplast only partially encircling the cell, usually with a single pyrenoid. Fragmentation common, as are thick, brown-walled akinetes.

It grows attached or free-floating in often soft, more acid water areas where desmids tend also to be abundant.

Most authors have chosen to maintain it as distinct from *Gloeotila*, a genus accommodating species having a strict contiguous arrangement of cells and lacking pyrenoids (Lokhorst, 1991). Several *Geminella* and/or *Gloeotila* species lacking pyrenoids and a mucilaginous sheath have been transferred by Hindák (1996) to the genus *Stichococcus*. It is traditionally placed in the order Ulotrichales, but is more closely related to the Chlorococcales based upon features associated with mitosis and cellular division (Hoek *et al.*, 1995).

Gloeotila Kützing (Fig. 4L)

Filaments of *Gloeotila* consist of loosely adherent cells with a tendency to break into short lengths, surrounded by a homogeneous mucilaginous sheath. Cells are generally longer than broad, oblong or ellipsoidal, and blunt at apices, each with a parietal, laminate or girdle-like chloroplast encircling most of the cell, pyrenoids are absent. Zoospores are biflagellate.

It grows attached, free-floating and planktonic, or on soil.

Sometimes it is considered congeneric with *Geminella*, but is usually separated on the absence of pyrenoids and its propensity to fragment. See comments under *Geminella*.

Gloeotilopsis Ramanathan (Fig. 4O)

Cells of *Gloeotilopsis* are solitary, rarely loosely attached, without a mucilaginous sheath. Cells are cylindrical and bluntly rounded at apices, each with a parietal chloroplast encircling most of the cell and a single pyrenoid.

The North American species is only known from a culture of soil from Dauphin Island, Alabama.

It is a little known and doubtful genus.

Hormidiopsis Heering (Fig. 4M)

Hormidiopsis filaments consist of a series of loosely attached mucilaginous sheaths; within each sheath lies an interrupted row of two or four cells. Cells are oval or oblong, shorter than broad, each possess a parietal chloroplast encircling part of the cell, with or without a single pyrenoid.

It is associated with soil and intermingled with algae in small boggy pools. Considerable doubt surrounds the two known species, with the type (*H. crenulatum*) a subaerial alga not reported to have a pyrenoid. The genus *Hormidiopsis* is possibly congeneric with *Ulothrix* or *Klebsormidium* (see remarks in Prescott, 1983; Bourrelly, 1988).

Klebsormidium Silva, Mattox *et* Blackwell (Fig. 4C)

Klebsormidium filaments are without differentiation into a basal or apical cell. Cells are cylindrical or beadlike (doliform), walls thin or thickened and often lamellated, if thickened then rough (crenulate and/or verrucose), sometimes cross walls are surrounded by H-shaped pieces, each with a parietal, elliptical, discoid or girdle-shaped chloroplast normally encircling less than 80% of the cell, typically with a single pyrenoid. Zoospores are biflagellate and produced in unspecialized cells and released through a pore. Aplanospores and thick-walled akinetes are produced.

Often one of the more abundant green filamentous algae in temperate streams, it is frequently regarded as a seasonal annual, sometimes growing intermingled with other filamentous forms. It is commonly subaerially on wet soil, dripping rocks, or seepage areas and when dry often developing thick H-shaped walls. Some species are especially abundant in acid environments where the water is contaminated by high concentrations of heavy metals.

Characters traditionally used to distinguish this species are known to be influenced by environmental conditions. According to Lokhorst (1996), for accurate and reliable identification it is desirable to examine material after growing under controlled laboratory conditions. About 15 species are known.

It is usually placed in the Order Klebsormidiales based on ultrastructural details (in older literature as *Hormidium*, but now invalid).

Koliella Hindák (Fig. 4J)

The cells of *Koliella* are solitary, usually detaching immediately after dividing and, only more rarely, daughter cells remain in pairs for a time to form short pseudofilaments before fragmenting. Cells are spindle-shaped, filiform, or very elongated, straight or curved, apices obtuse, rounded, or gradually tapering to form very long points, each contain a parietal, laminate or girdle-shaped, straight, or spiral chloroplast, with or without pyrenoids and oil droplets present. Zoospores are biflagellate. Sexual reproduction is oogamous, with details uncertain. Akinetes are produced.

Usually considered to be widely distributed in freshwater plankton, it is occasionally periphytic, epilithic or associated with ice and snow banks. The only North American record traced so far is a mention of "*Koliella* sp." by Prescott (1983) based upon material collected from snow banks in the Olympic Mountains, Washington.

It is very similar to *Raphidonema*, the two possibly being congeneric.

Koliella is placed in the Order Klebsormidiales based upon details of its ultrastructure.

Pseudoschizomeris Deason *et* H. C. Bold (Fig. 4P)

Pseudoschizomeris filaments are unbranched in young cultures; with age it develops distally more than one series of spherical cells, eventually producing much mucilage and becoming palmelloid. Cells each possess a parietal chloroplast encircling most of the cell and a single pyrenoid.

It is only known from a culture of Texas soil and is monospecific.

Radiofilum Schmidle (Fig. 4G)

Radiofilum filaments are usually unbranched but occasionally falsely branched, consisting of a row of cells whose wall is of two helmet-shaped halves with a ringlike transverse rim formed where they join, and surrounded by a thick mucilaginous sheath. Cells are spherical, subspherical, or ellipsoidal, each possess a parietal, cup-shaped chloroplast lying adjacent to the cross wall, with a single pyrenoid. Reproduction is by fragmentation.

It occurs in various aquatic habitats including pools and ponds. Three species are known to be widely distributed in the United States.

It is traditionally assigned to the order Ulotrichales, but placed in the Chlorococcales by Hoek *et al.* (1995).

Raphidonema Lagerheim (Fig. 4K)

The filaments of *Raphidonema* are two to 32 cells in length. Cells are cylindrical, only apical cells acuminate or gradually tapering to a point, walls thin, each containing a parietal, laminate or girdle-shaped chloroplast, with or without pyrenoids. Zoospores are biflagellate. Sexual reproduction is oogamous: male gametes biflagellate (one per cell) and vegetative cells functions as a female gamete.

Some species are found in aquatic habitats and others are associated with snow or ice, the latter of which become green when the alga is present in quantity. About 20 species are recorded.

It is placed in the Order Klebsormidiales based upon ultrastructural details.

Raphidonemopsis Deason (Fig. 4N)

The cells of *Raphidonemopsis* are solitary or form two-celled filaments, with a small attachment disk and are terminally bluntly pointed. Cells gradually narrow to the base and apex, walls thin, each containing a parietal, laminate chloroplast and a single pyrenoid.

It is known only from soil enrichment cultures.

This is a doubtful genus which is principally distinguished from *Raphidonema* by its attachment to a surface.

Stichococcus Nägeli (Fig. 4D)

Cells are solitary or form a readily dissociating filament, without a mucilaginous sheath. Cells cylindrical and often elongated with rounded apices, sometimes short, spherical to slightly oval, thin-walled, each containing a single parietal, laminate chloroplast often covering only a small part of the cell, normally without a pyrenoid. Reproduction is by fragmentation and cell division.

It has a very wide ecological amplitude, ranging from soil-free surfaces to freshwater and marine habitats, although it is most common on tree trunks, wooden fencing, or damp soil, and sometimes associated with other subaerial green algae (e.g., *Desmococcus* and *Apatococcus*).

It is placed in the Order Klebsormidiales based on ultrastructural details.

Ulothrix Kützing (Fig. 4F)

The filaments of *Ulothrix* are attached by a single basal cell or rhizoidal basal cell, or by rhizoids arising from other cells. Cells are cylindrical, sometimes bead-like (doliiform), often longer than broad, walls thin or thick and occasionally lamellated or roughened, with "H-shaped" pieces, each has a parietal, girdle-shaped, and more usually a lobed chloroplast, encircling all or normally over three quarters of the cell, with one or more pyrenoids. Zoospores are quadriflagellate. Gametes are biflagellate and isogamous. Aplanospores and thick-walled akinetes are produced. Unicellular stage in life history, the *Codiolum*-stage.

It grows attached or free-living on damp soil, cliffs wetted by spray, as well as in stagnant water; it is less common in rapidly flowing water or on stones and wood along the margins of wave-splashed lakeside shores, with the possible exception of the Great Lakes.

Uronema Lagerheim (Fig. 4A)

Uronema filaments are attached by a basal cell, often terminating in a slightly curved and asymmetrically pointed apex. Cells are usually cylindrical, each with a parietal, laminate chloroplast partially encircling the cell, one to four pyrenoids. Zoospores are quadriflagellate, one or two per sporangium.

Most grow epiphytically on aquatic plants in various freshwater bodies. About 10 species have been described.

Taxonomic position of this little-known genus remains uncertain. It is retained in the order to which it has been traditionally assigned, but ultrastructural features of cell division and the basal body of flagellated stages suggest that it should be assigned to the Chaetophorales *sensu stricto* (Hoek *et al*., 1995).

Order Ulvales

Order Ulvales is membrane-like or tubular and normally branched, single-layered, and parenchymatous. Cells each possess a single nucleus, a parietal laminate chloroplast and one to several pyrenoids.

Enteromorpha Link (Fig. 7A)

Enteromorpha is macroscopic, consisting of attached elongated, tubular sacks, usually branched and sometimes with numerous proliferations, initially attached by a basal rhizoidal cell or proliferations developing from its base. Cells are quadrate or cylindrical, often angular by mutual compression, each possess a parietal, laminate chloroplast, and one or several pyrenoids. Zoospores are quadriflagellate. Sexual reproduction is by biflagellate gametes.

The genus grows often initially attached when growing on submerged objects in flowing-water or occasionally in ponds, it frequently becomes free-floating, inflated, and intestiniform (intestine-like). Only one species (*Enteromorpha flexuosa*) is considered to be freshwater, with many others marine and brackish-water. Many North American inland records (e.g., Silver Springs, Wyoming County, western New York State; see Catling and McKay, 1980; salt lakes in Kansas; Ranch, 1981) are of species usually reported from marine habitats (*Enteromorpha intestinalis* and *E. prolifera*). These plants were collected growing on rocks in salt springs or in streams and lagoons receiving effluent from salt mines.

Monostroma Thuret (Fig. 7D)

Monostroma is macroscopic, initially sacklike, and later splitting to form a single-layered membrane, parenchymatous or cells rounded and grouped in fours or separated by mucilage, commonly attached by rhizoidal protuberances. Cells are angular by compression or rounded, each with a single, parietal chloroplast encircling most of the cell and a single pyrenoid. Reproduction is by quadriflagellate zoospores and biflagellate anisogametes. *Codiolum*-stage in life history. Various types of life history are known in genus.

Only one or two species are known exclusively from freshwater habitats, otherwise it is a marine and brackish-water genus. It often grows on rocks or lodged driftwood in swift-flowing streams and rivers (Taft, 1964); it is also reported from standing water in Arctic Canada.

It is retained in the Ulvales where it is traditionally placed, but Hoek *et al*. (1995) consider it to be a multicellular, thalloid member of the Order Codiolales (= Ulotrichales) based on ultrastructural features associated with two marine species so far examined.

VII. GUIDE TO LITERATURE FOR SPECIES IDENTIFICATION

The majority of freshwater algal genera have a world-wide distribution. For this reason, floristic works covering other regions of the world can be used in identifying North American algae. Many of the more useful and comprehensive works dealing with green algae are frequently in languages other than English. For example, Starmach's series of volumes on the freshwater algae of Poland are very useful since coverage is world-wide. These are written in Polish, although the volume dealing with filamentous green algae (Starmach, 1972) has a key in English. Very few libraries or institutions hold more than a fraction of these works, some of which are difficult to use even by specialists possessing a working knowledge of the language. Most of the most useful monographs, flora's and identification guides on freshwater algae are listed in "Key Works to the Fauna and Flora of the British Isles and North-western Europe" (Sims *et al*., 1988).

The majority of the green algal genera considered in this chapter are mentioned in Smith's classical work *Fresh-water Algae of the United States* (Smith, 1933, 1950) and various regional floristic works (e.g., Prescott, 1951; Tiffany and Britton, 1952; Whitford and Schumacher, 1963). The regional floristic works are useful for species level identification since they contain descriptions, keys, and are profusely illustrated. Sometimes the charophytes are not included in these treatments, one of the few major algal groups for which there exists a complete world-wide review. This review by Wood and Imahori (1964, 1965) still remains the most comprehensive treatment of the group and includes much data on North American charophytes. The first author also published a useful guide to the charophytes of North America, Central America, and the West Indies (Wood, 1967) and 20 years earlier had reviewed the genus *Nitella* in North America (Wood, 1948). One of the first comprehensive treatments of North American green filamentous algae was Hazen's monograph *The Ulothricaceae and Chaetophoraceae of the United States* (Hazen, 1902) with many considered in *The Green Algae of North America* by Collins (1904). Another monographic treatment was Tiffany's monumental work on North American Oedogoniales (Tiffany, 1937) following an earlier publication dealing with the order world-wide (Tiffany, 1930). Unfortunately, all these useful identification works are unobtainable except through secondhand booksellers. The most useful modern series of identification guides for North American algae has been written by Dillard and covers the southeastern part of the United States. The volume dealing with the filamentous green algae was published in 1989 and like others in the series mentions taxa known elsewhere in the United States. The most useful and comprehensive volume on filamentous and parenchymatous freshwater green algae is undoubtedly *Die Chaetophoralean der Binnengewässer* (Printz, 1964). Like the major of works mentioned above, it has been long out of print and only obtainable through the secondhand book trade.

The following is a list of works dealing with North American genera; often these cite references to earlier literature. Sometimes no modern treatment exists, and the most comprehensive account is to be found in earlier, sometimes obscure, and often out of print works (e.g., Printz, 1964; Smith, 1950; Starmach, 1972). Otherwise, identification has to be based on keys, descriptions, and illustrations in more general floristic works such as John *et al*., (2002).

1. *Apatococcus*—Ettl and Gartner (1995), Gartner and Ingolic (1989), Prescott (1983), Printz (1964).
2. *Aphanochaete*—Dillard (1989), Prescott (1962), Printz (1984), Starmach (1972), Tupa (1974).
3. *Basicladia*—Dillard (1989), Prescottt (1983).
4. *Binuclearia*—Dillard (1989), Prescott (1983).
5. *Bulbochaete*—Dillard (1989), Mrozinska (1985), Tiffany (1930, 1937, 1944).
6. *Cephaleuros*—Dillard (1989), Printz (1964), Thompson and Wujek (1997).
7. *Chaetomorpha*—Faridi (1962).
8. *Chaetonema*—Dillard (1989), Prescott (1983), Printz (1964), Tupa (1974).
9. *Chaetophora*—Dillard (1989), Prescott (1983), Printz (1964), Stamarch (1972).
10. *Chaetosphaeridium*—Dillard (1989), Prescott (1983), Printz (1964).
11. *Chamaetrichon*—Tupa (1974).
12. *Chara*—Wood (1967), Wood and Imahori (1964, 1965).
13. *Chlorochytrium*—Chapman and Waters (1992), Prescott (1983).
14. *Chlorotylium*—Dillard (1989), Prescott (1983), Printz (1964).
15. *Cladophora*—Hoek (1963, 1982).
16. *Cloniophora*—Dillard (1989), Islam (1961), Prescott (1983), Starmach (1972).
17. *Coleochaete*—Dillard (1989), Prescott (1983), Printz (1964), Szymanska and Spalik (1993).
18. *Conochaete*—Prescott (1983), Printz (1964), Smith (1933, 1950).
19. *Ctenocladus*—Blinn and Stein (1970), Prescott (1983).
20. *Cylindrocapsa*—Dillard (1989).
21. *Dermatophyton*—Prescott (1983), Printz (1964), Smith (1933, 1950), Starmach (1972).
22. *Desmococcus*—Ettl and Gartner (1995), Gartner and Ingolic (1989), Printz (1964).
23. *Dicranochaete*—Dillard (1989), Matula (1992), Prescott (1983), Printz (1964).
24. *Dichotomosiphon*—Dillard (1989), Prescott (1983).
25. *Dilabifilum*—Johnson and John (1990).
26. *Draparnaldia*—Dillard (1989), Forest (1976), Prescott (1983), Printz (1964), Starmach (1972).
27. *Draparnaldiopsis*—Forest (1976), Printz (1964), Prescott (1983), Starmach (1972).

28. *Entocladia*—Dillard (1989), Prescott (1983), Printz (1964), Smith (1933, 1959).
29. *Elakatothrix*—Prescott (1983), Printz (1964).
30. *Entermorpha*—Prescott (1983), Bliding (1963).
31. *Filoprotococcus* (=*Trichosarcina)*—Nichols and Bold (1965), Prescott (1983), Thompson and Wujek (1996).
32. *Fridaea*—Prescott (1983), Printz (1964).
33. *Fritschiella*—Prescott (1983), Printz (1964).
34. *Geminella*—Dillard (1989), Prescott (1983).
35. *Gloeotila*—Printz (1964), Ramanthan (1964), Smith (1933, 1950).
36 *Gloeotilopsis*—Ramanthan (1964).
37. *Gomontia*—Dillard (1989), Prescott (1983).
38. *Gongrosira*—Dillard (1989), Printz (1964), Starmach (1972), Tupa (1974).
39. *Hazenia*—Bold (1958), Prescott (1983).
40. *Helicodictyon*—Dillard (1989), Biebl (1968), Prescott (1983), Whitford and Schumacher (1969).
41. *Hormidiopsis*—Prescott (1983).
42. *Klebsormidium*—Dillard (1989), Ettl and Gartner (1995), Lokhorst (1996), Ramanthan (1967).
43. *Koliella*—Hindak (1983), Prescott (1983).
44. *Leptosira*—Dillard (1989), Ettl and Gartner (1995), Starmach (1972), Steil (1944), Tupa (1974), Wujek (1971).
45. *Microspora*—Dillard (1989), Prescott (1983), Lokhorst (1999), Ramanthan (1964).
46. *Microthamnion*—Dillard (1989), John and Johnson (1987), Printz (1964).
47. *Monostroma*—Bliding (1968), Prescott (1983), Taft (1964).
48. *Nitella*—Wood (1948, 1967), Wood and Imahori (1964, 1965).
49. *Oedocladium*—Dillard (1989), Ettl and Gartner (1995), Mrozinska (1985), Tiffany (1930, 1937).
50. *Oedogonium*—Dillard (1989), Hoffman (1967), Tiffany (1930, 1937), Yung *et al.* (1986).
51. *Oligochaetophora*—Prescott (1983), Printz (1964).
52. *Phycopeltis*—Printz (1994), Prescott (1983).
53. *Phyllosiphon*—Dillard (1989), Prescott (1983).
54. *Pithophora*—Dillard (1989), Prescott (1983).
55. *Pleurastrum*—Groover and Bold (1969), Starmach (1972), Tupa (1974).
56. *Polychaetophora*—Prescott (1983), Smith (1933, 1950).
57. *Prasiola*—Ettl and Gartner (1995), Rindi *et al.* (1999), Smith (1933, 1950).
58. *Printzia*—Thompson and Wujek (1992).
59. *Protoderma*—Dillard (1989), Printz (1964), Starmach (1972), Tupa (1974).
60. *Protosiphon*—Dillard (1989), Whitford and Schumacher (1967).
61. *Pseudendoclonium*—Dillard (1989), John and Johnson (1989), Printz (1964), Tupa (1974).
62. *Pseudochaete*—Prescott (1983), Smith (1933, 1950).
63. *Pseudoschizomeris*—Deason and Bold (1960), Ettl and Gartner (1995), Prescott (1983).
64. *Pseudulvella*—Dillard (1989), Prescott (1983), Printz (1964).
65. *Radiofilum*—Dillard (1989), Prescott (1983), Printz (1964).
66. *Raphidonema*—Dillard (1989), Hindák (1963), Hoham (1973), Prescott (1983), Printz (1964).
67. *Raphidonemopsis*—Deason (1969), Ettl and Gartner (1995).
68. *Rosenvingiella*—Edwards (1975).
69. *Rhizoclonium*—Dillard (1989), Prescott (1983).
70. *Schizomeris*—Campbell and Sarafis (1972), Dillard (1989), Prescott (1983), Printz (1964).
71. *Sphaeroplea*—Dillard (1989), Prescott (1983).
72. *Stichococcus* – Dillard (1989), Ettl and Gartner (1995), Hindák (1996), Printz (1964), Ramanthan (1964).
73. *Stigeoclonium*—Dillard (1989), Cox and Bold (1966), Islam (1963), Printz (1964), Starmach (1974).
74. *Stomatochroon*—Thompson and Wujek (1997).
75. *Thamniochaete*—Dillard (1989), Prescott (1983), Printz (1964), Starmach (1974).
76. *Tolypella*—Wood (1946, 1967), Wood and Imahori (1964, 1965).
77. *Trentepohlia*—Dillard (1989), Ettl and Gartner (1995), Printz (1964), Starmach (1972).
78. *Trichodiscus*—Printz (1964), Starmach (1974).
79. *Trichophilus*—Bourrelly (1988), Printz (1964), Starmach (1974).
80. *Ulothrix*—Dillard (1989), Lokhorst (1979), Printz (1964), Starmach (1972).
81. *Uronema*—Dillard (1989), Ettl and Gartner (1995), Prescott (1983), Printz (1964), Starmach (1972).

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